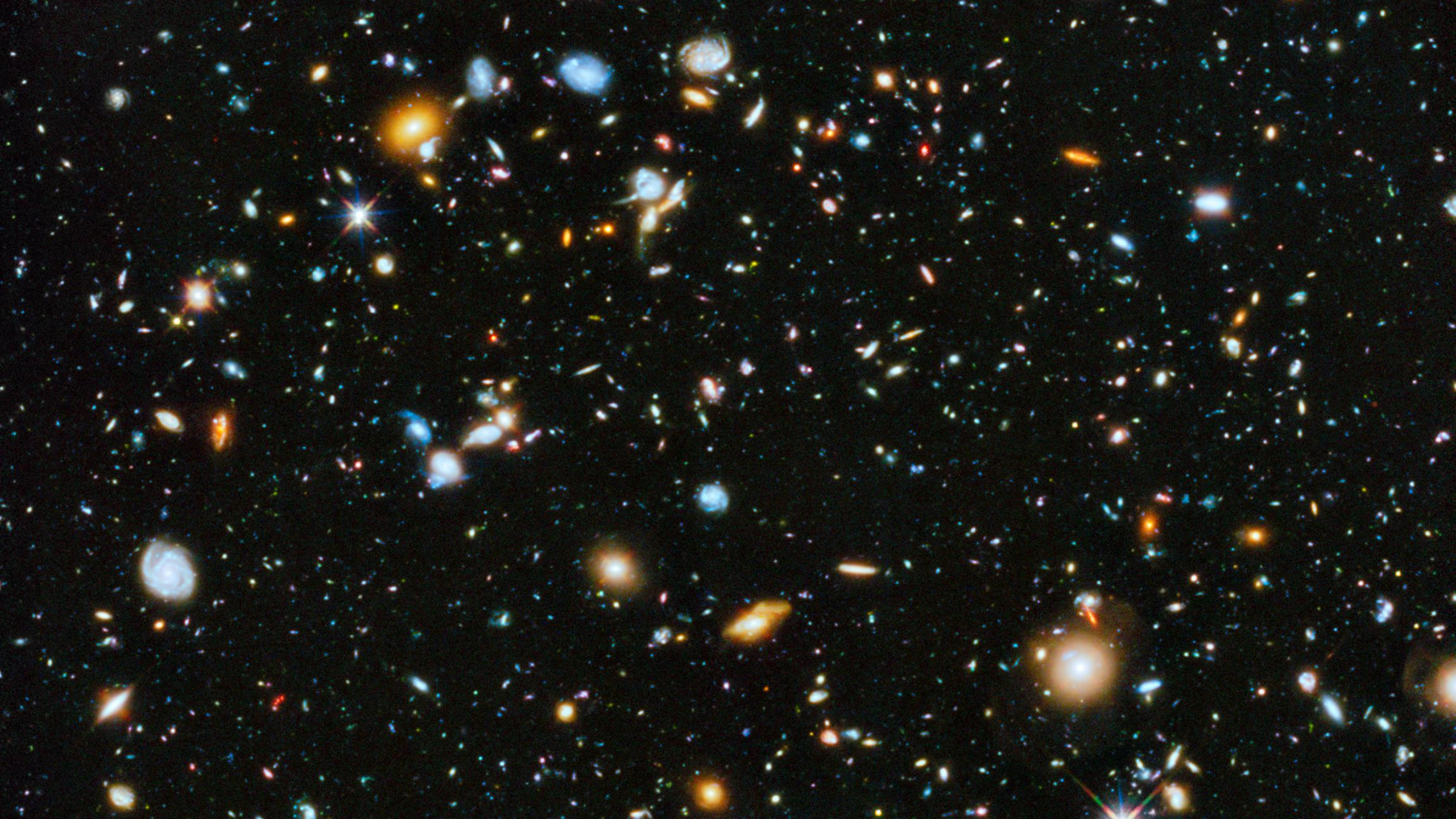


RECORDING





HUBBLE UNVEILS ITS MOST COLORFUL VIEW OF THE UNIVERSE
(ZOOM AND PAN)

BROWSE THE LINK BELOW AND PLAY!

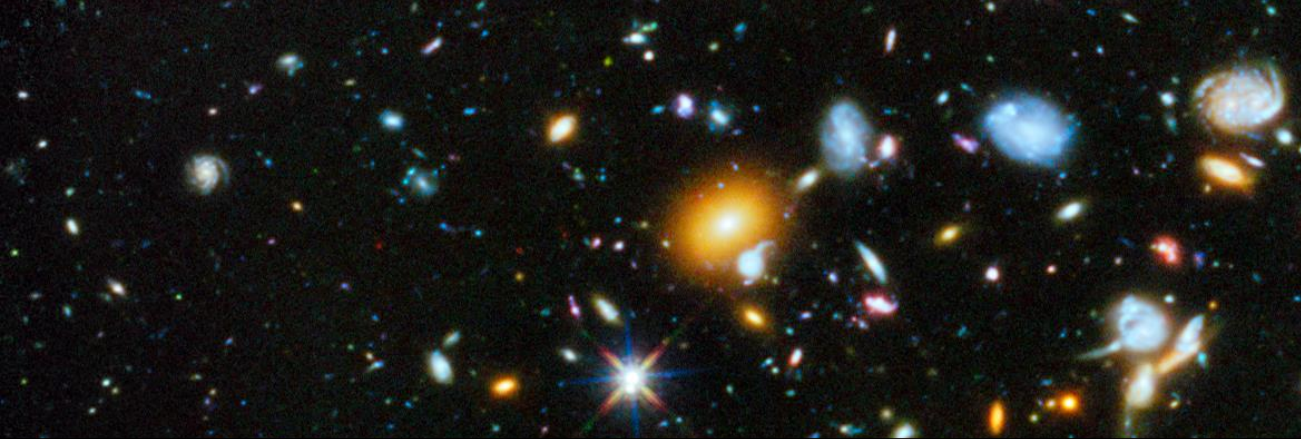
<https://hubblesite.org/contents/media/videos/2014/27/766-Video.html>



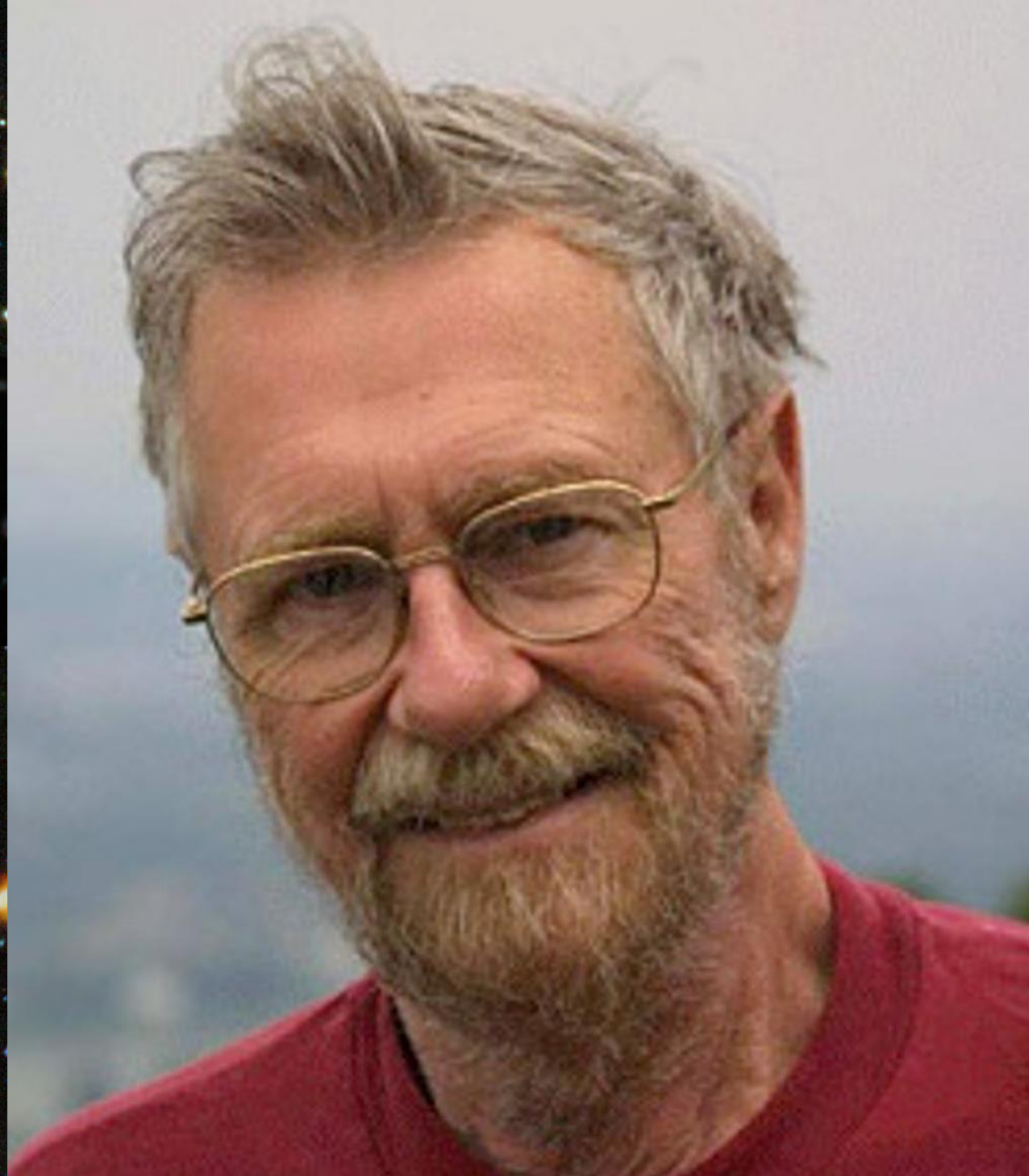
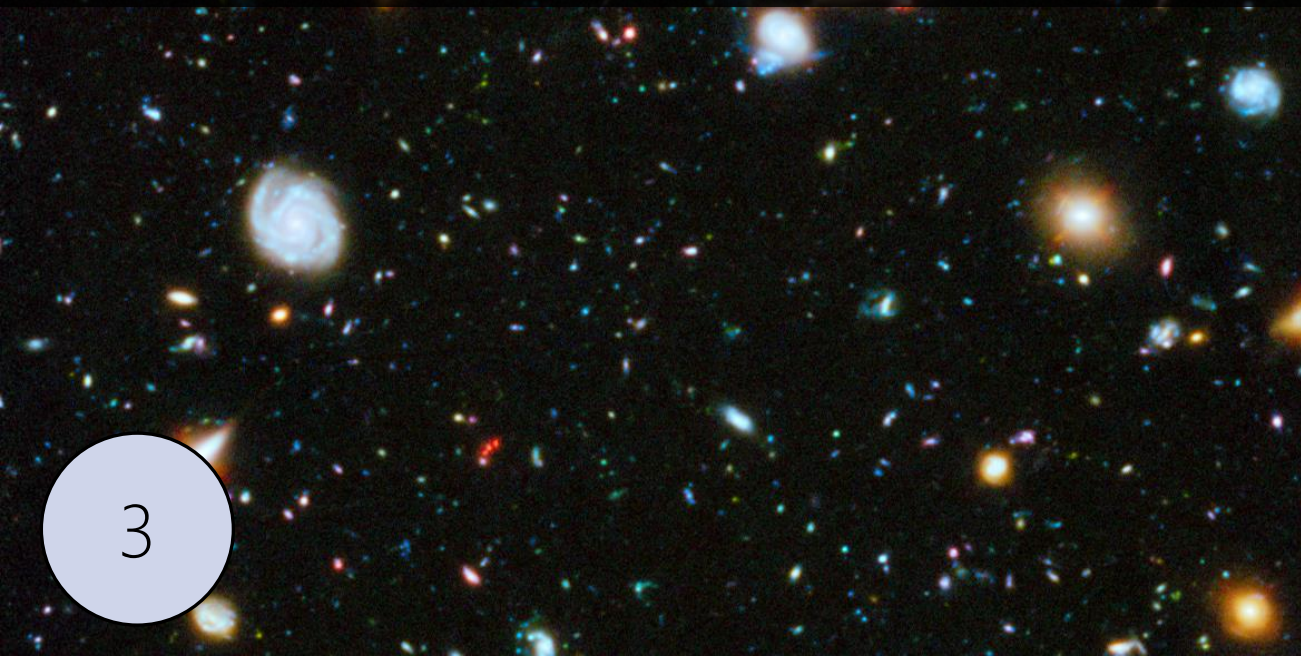
COMPUTER SCIENCE & COMPUTERS
ASTRONOMY & TELESCOPES

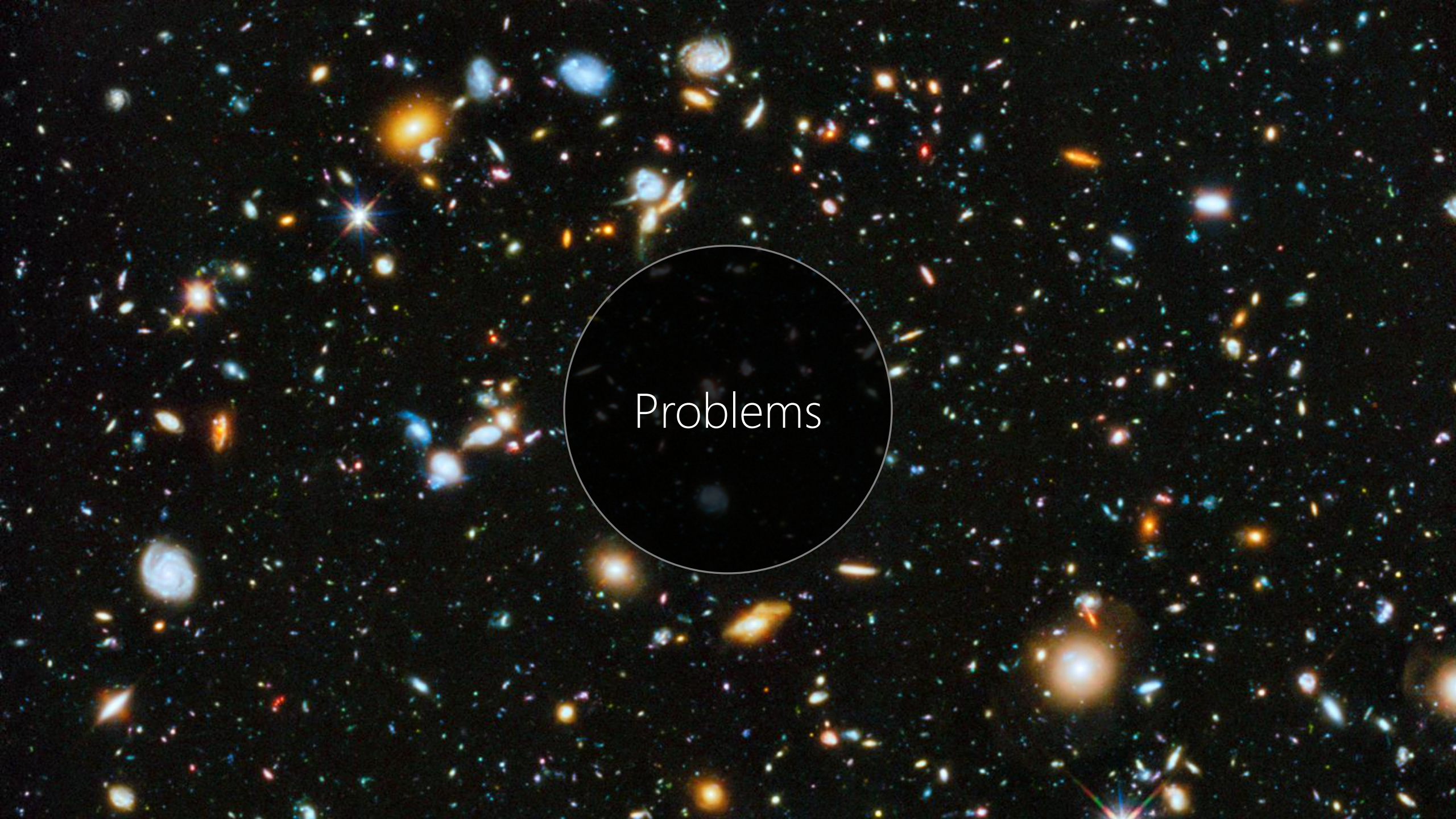


ZOOM INTO CPU AND MEMORY
ZOOM INTO UNIVERSE AND STARS

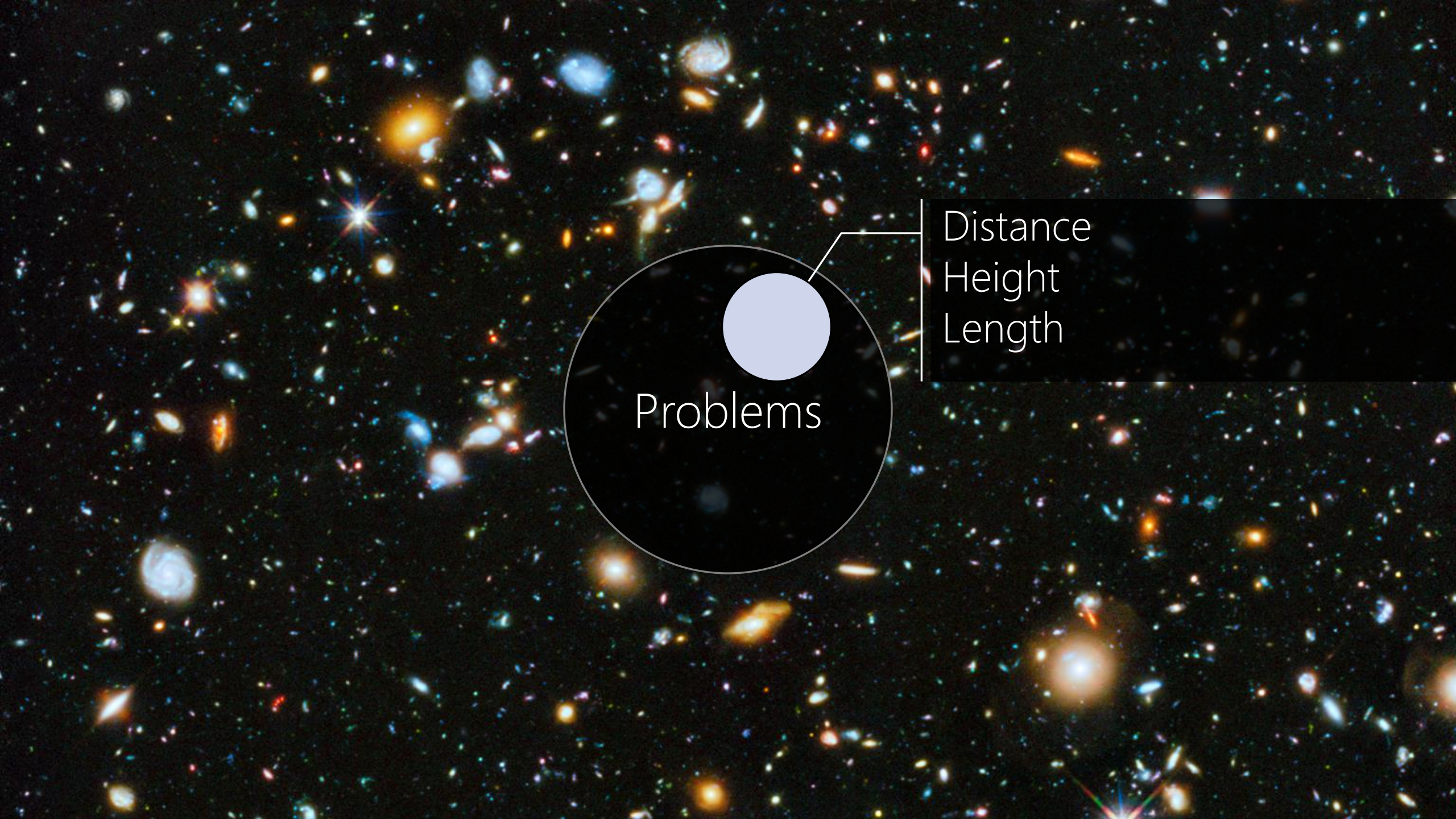


"Computer science is no more about computers than astronomy is about telescopes" - Edsger Dijkstra



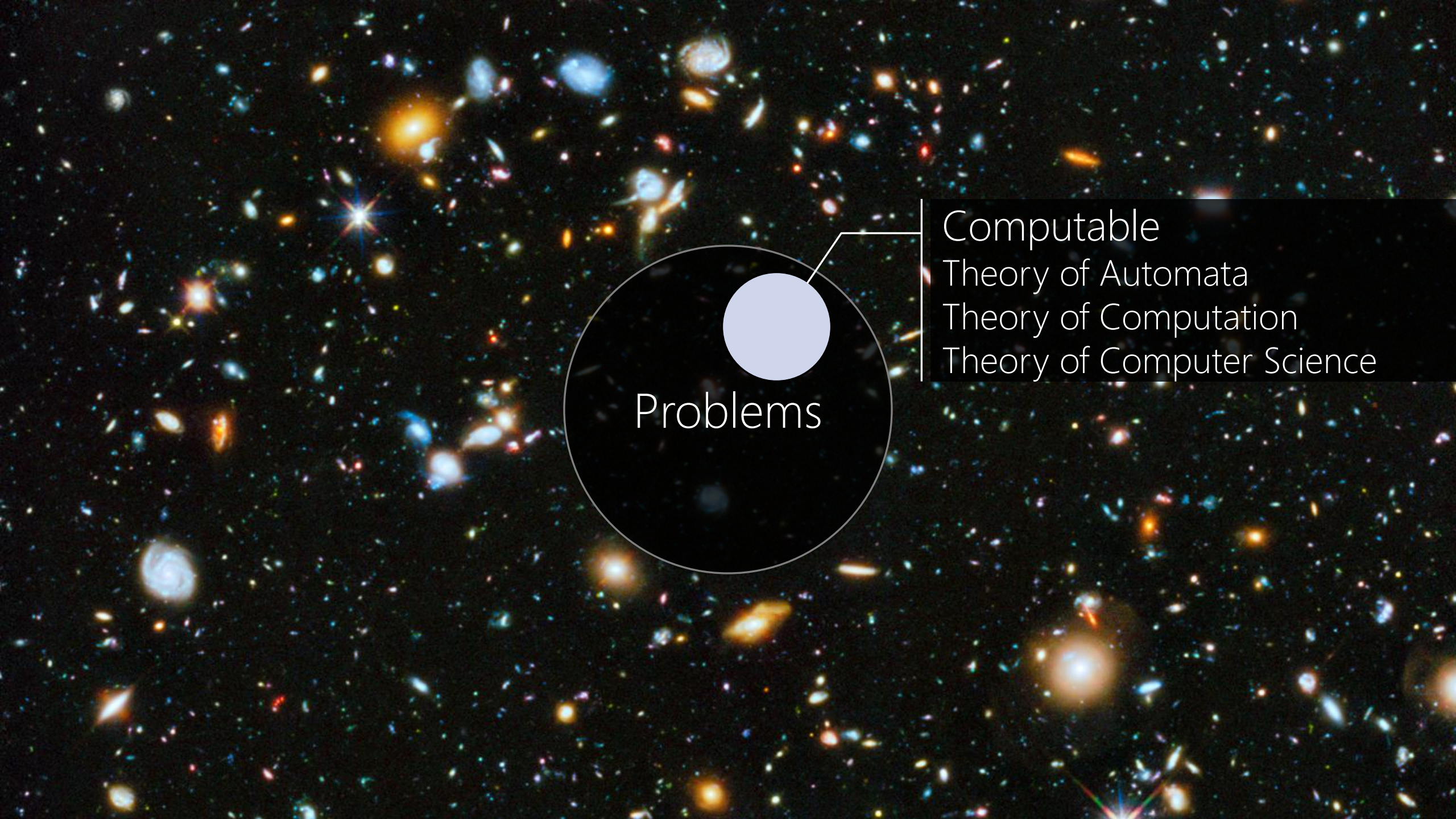
A deep-field astronomical image showing a vast number of galaxies in various colors (blue, orange, red, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, diffuse clouds and others as more compact, point-like sources. A central black circle with a thin white border contains the word "Problems" in white text.

Problems



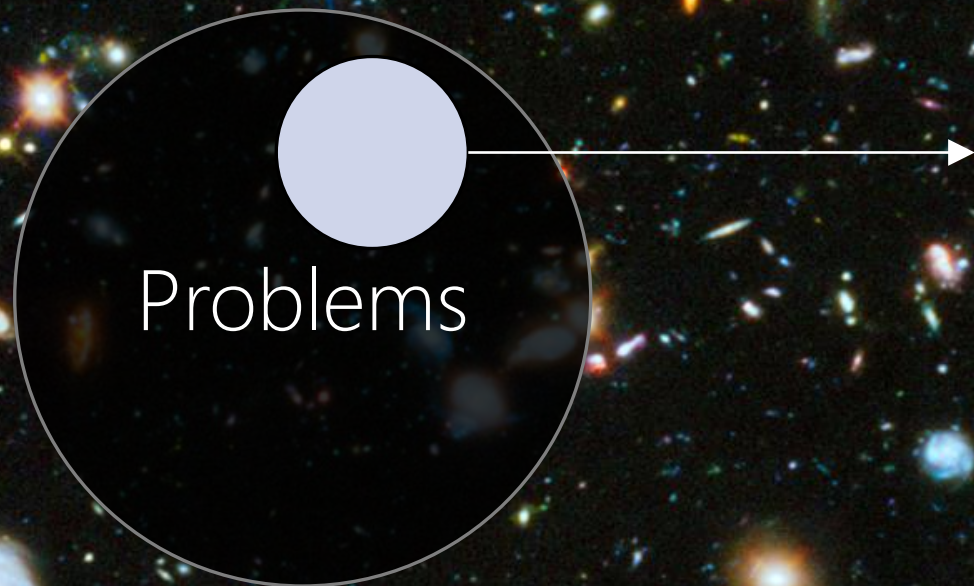
Problems

Distance
Height
Length

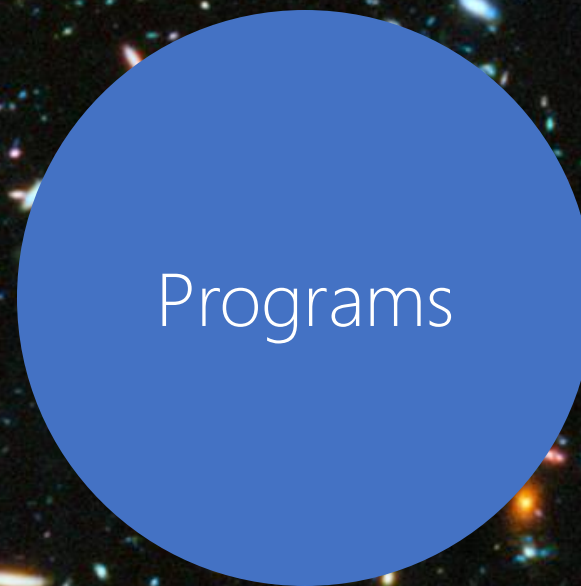
The background is a vibrant deep space photograph showing a dense cluster of galaxies. The galaxies are in various stages of evolution, with some appearing as bright, yellowish-orange elliptical shapes and others as more complex, blue-tinted structures. A large, dark grey circle is centered in the lower half of the image. Inside this circle, at the top, is a smaller, solid light blue circle. A thin white line extends from the right side of this light blue circle, pointing towards a dark grey rectangular box on the right edge of the image. The word "Problems" is written in white, sans-serif font, centered within the large dark grey circle.

Problems

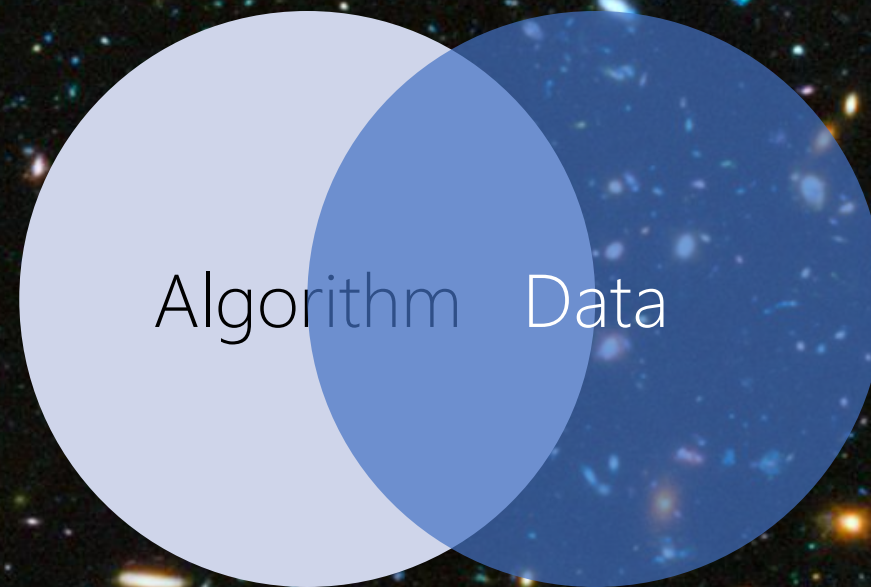
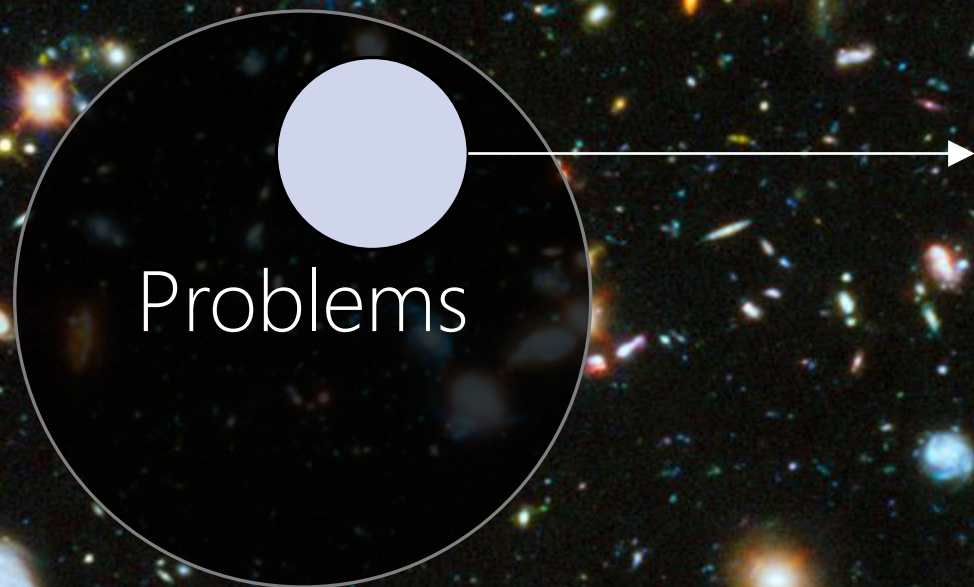
Computable
Theory of Automata
Theory of Computation
Theory of Computer Science



Problems



Programs



Algorithm Design
Algorithm Analysis
Artificial Intelligence (AI)
Machine Learning
Data Mining



Algo Data

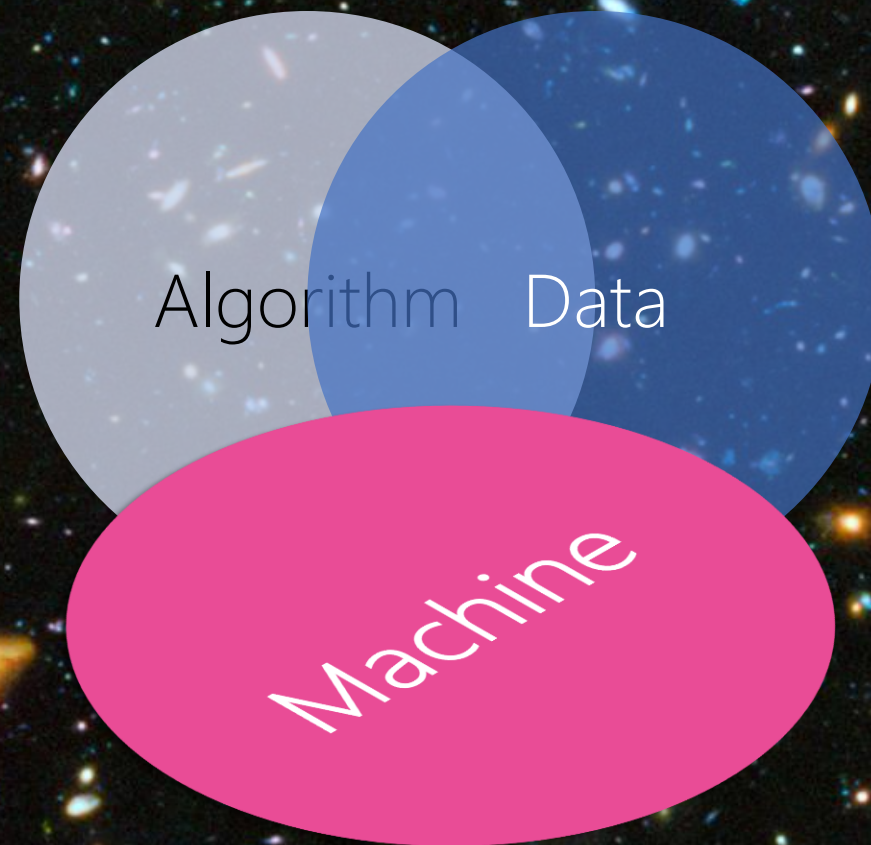
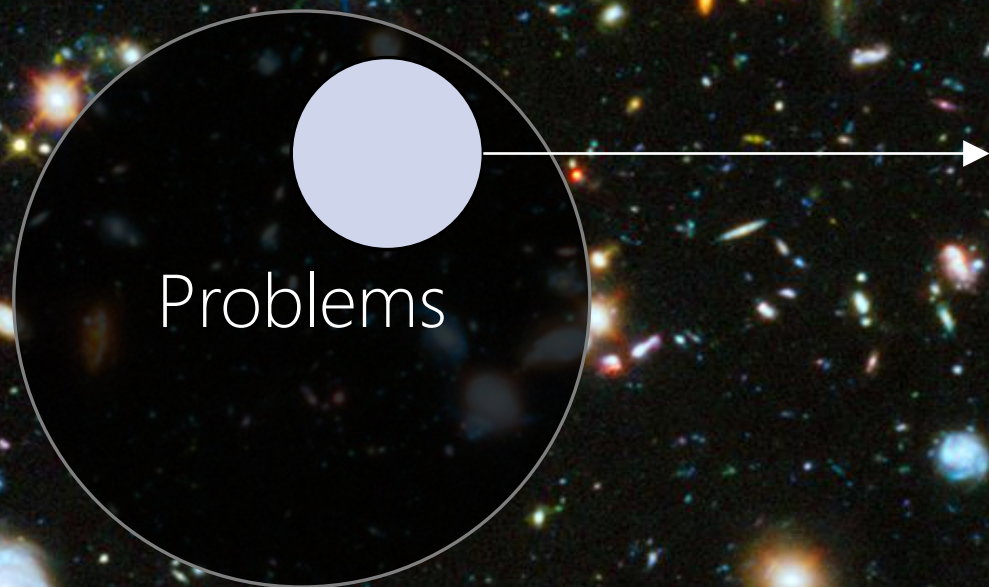
Data Structure (Memory)
File Structure (File Systems)
Database Management Sys.
Data Warehouse
Big Data
Cloud



Euclid – 300 BCE
Greatest Common Divisor

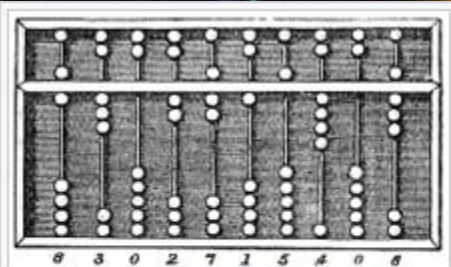
Archimedes – 212 BCE
Approximate π

Khwārizmī – 850 AD
Linear and Quadratic Equations





The Ishango bone, a bone tool dating back to prehistoric Africa.



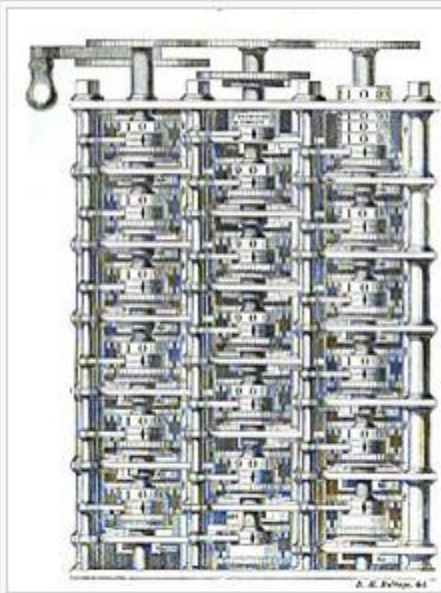
The Chinese **suanpan** (算盘). The number represented on this **abacus** is 6,302,715,408.

2700 – 2300 BC



The **Antikythera mechanism**, dating back to **ancient Greece** circa 150–100 BC, is an early **analog computing device**.

100 BC: to calculate astronomical positions



A portion of **Babbage's Difference engine**.

1833: Father of Computer



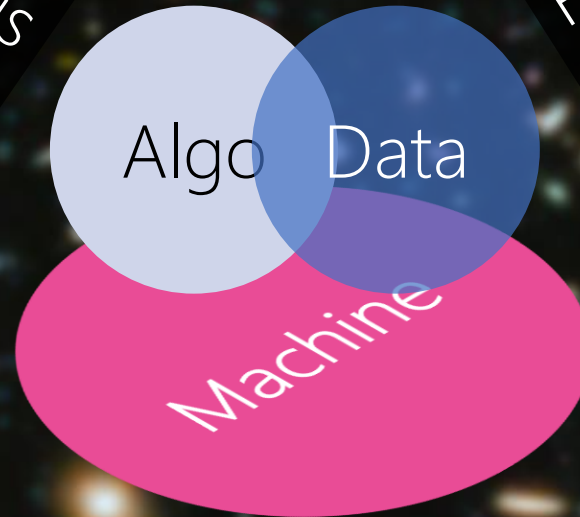
Replica of **Zuse's Z3**, the first fully automatic, digital (electromechanical) computer.

1941



Colossus, the first electronic digital **programmable** computing device, was used to break German ciphers during World War II. It is seen here in use at Bletchley Park in 1943.

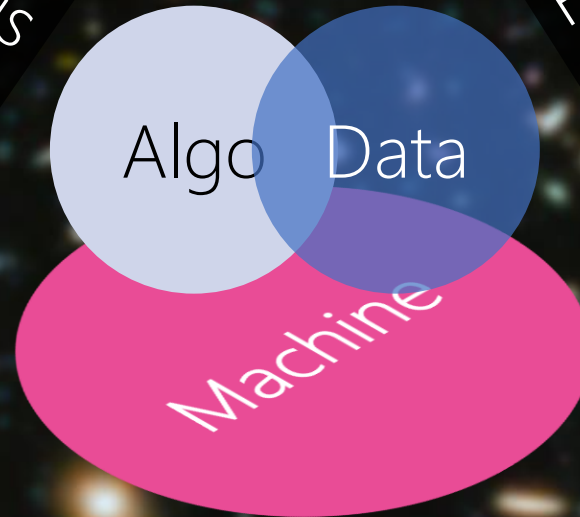
Algorithm Design
Algorithm Analysis
Artificial Intelligence (AI)
Machine Learning
Data Mining



Data Structure
File Structure
Database Management Sys.
Data Warehouse
Big Data
Cloud

Digital Design (Logic Circuits)
Computer Architecture
Microprocessor Programming
Assembly Language
Operating Systems

Algorithm Design
Algorithm Analysis
Artificial Intelligence (AI)
Machine Learning
Data Mining



Data Structure
File Structure
Database Management Sys.
Data Warehouse
Big Data
Cloud

No Need to Study Machines!

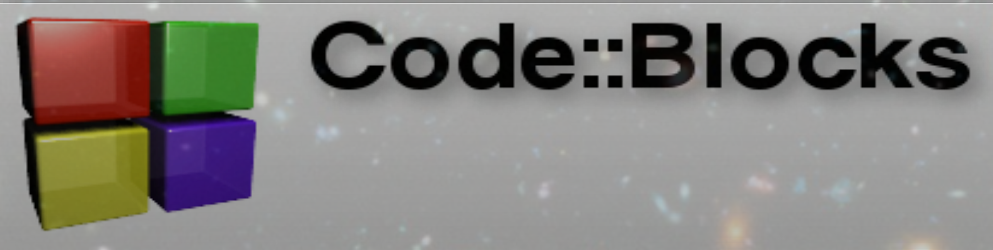
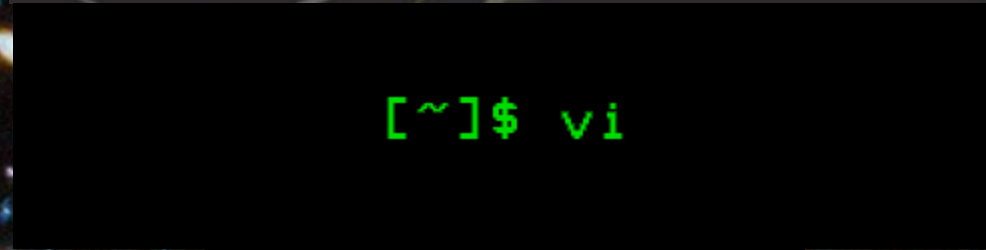


ARTIFICIAL
INTELLIGENCE



ChatGPT





```
1 #include <stdio.h>
2 int Add(int a, int b){
3     int result = a + b;
4     return result;
5 }
6 int main() {
7
8     int number1;
9     int number2;
10    printf("Enter two numbers:\n");
11
12    scanf("%d", &number1);
13    scanf("%d", &number2);
14
15    int sum = Add(number1, number2);
16
17    printf("%d + %d = %d", number1, number2, sum);
18
19    return 0;
20 }
```

Enter two integers:

12

35

$12 + 35 = 47$



Enter two integers:

999999

1

$999999 + 1 = 1000000$



Enter two integers:
9999999999999999
1



Enter two integers:

9999999999999999

1

-1530494977 + 1 = -1530494976



```
1 #include <stdio.h>
2 #include <limits.h>
3 int Add(int a, int b){
4     int result = a + b;
5     return result;
6 }
7 int main() {
8
9     int number1;
10    int number2;
11    printf("Enter two numbers between %d and %d:\n", INT_MIN, INT_MAX);
12
13    scanf("%d", &number1);
14    scanf("%d", &number2);
15
16    int sum = Add(number1, number2);
17
18    printf("%d + %d = %d", number1, number2, sum);
19
20    return 0;
21 }
22
```

Enter two integers between -2147483648 and 2147483647:

12

15

$12 + 15 = 27$



Enter two integers between -2147483648 and 2147483647:
2147483647
1



Enter two integers between -2147483648 and 2147483647:

2147483647

1

2147483647 + 1 = -2147483648



Enter two integers between -2147483648 and 2147483647:

2147483647

2

$2147483647 + 2 = -2147483647$



1 //Chitra Nayal, Mithun Kumar, Shivam Kushik, 04-06-2019, Sum of two large numbers
2 //Retrieved from <https://www.geeksforgeeks.org/sum-two-large-numbers/>

```
#include <iostream>
using namespace std;

string add(string a, string b) {
    //
}

int main() {
    //
}
```

```
22     int carry = 0;
23     for (int i = 0; i < n1; i++) {
24         // Do school mathematics, compute sum of current digits and carry
25         int sum = ((a[i] - '0') + (b[i] - '0') + carry);
26         result.push_back(sum % 10 + '0');
27         // Calculate carry for next step
28         carry = sum / 10;
29     }
```

Enter two integers:

2147483647

1

$2147483647 + 1 = 2147483648$

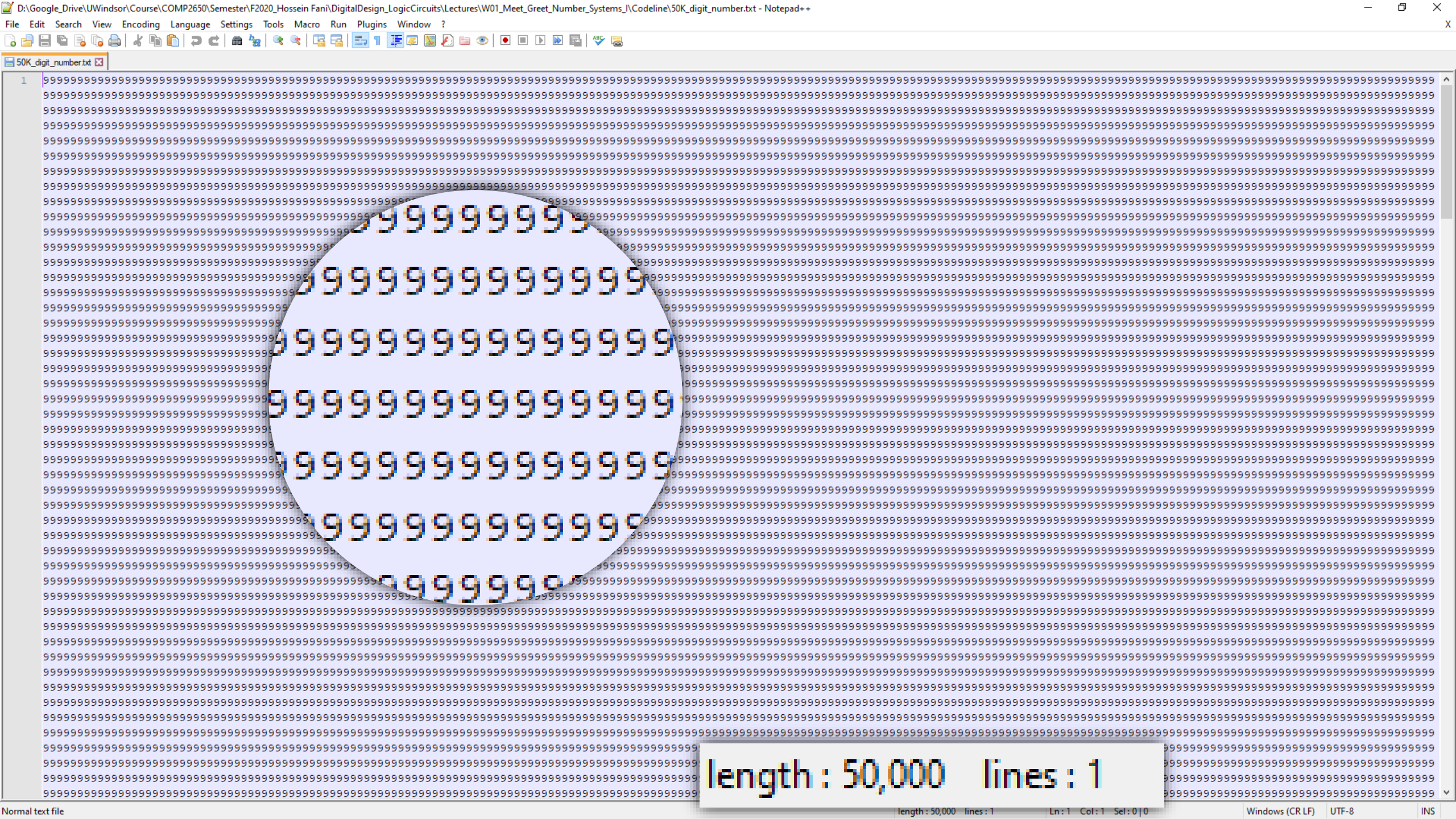


Enter two integers:

999

1

[illegible]



Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
999...99999 + 999...99999 = ?



Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
999...99999 + 999...99999 = 1999 ... 9998 => 50,001 digits



Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
999...99999 + 999...99999 = 1999 ... 9998 => 50,001 digits
Total time = ?

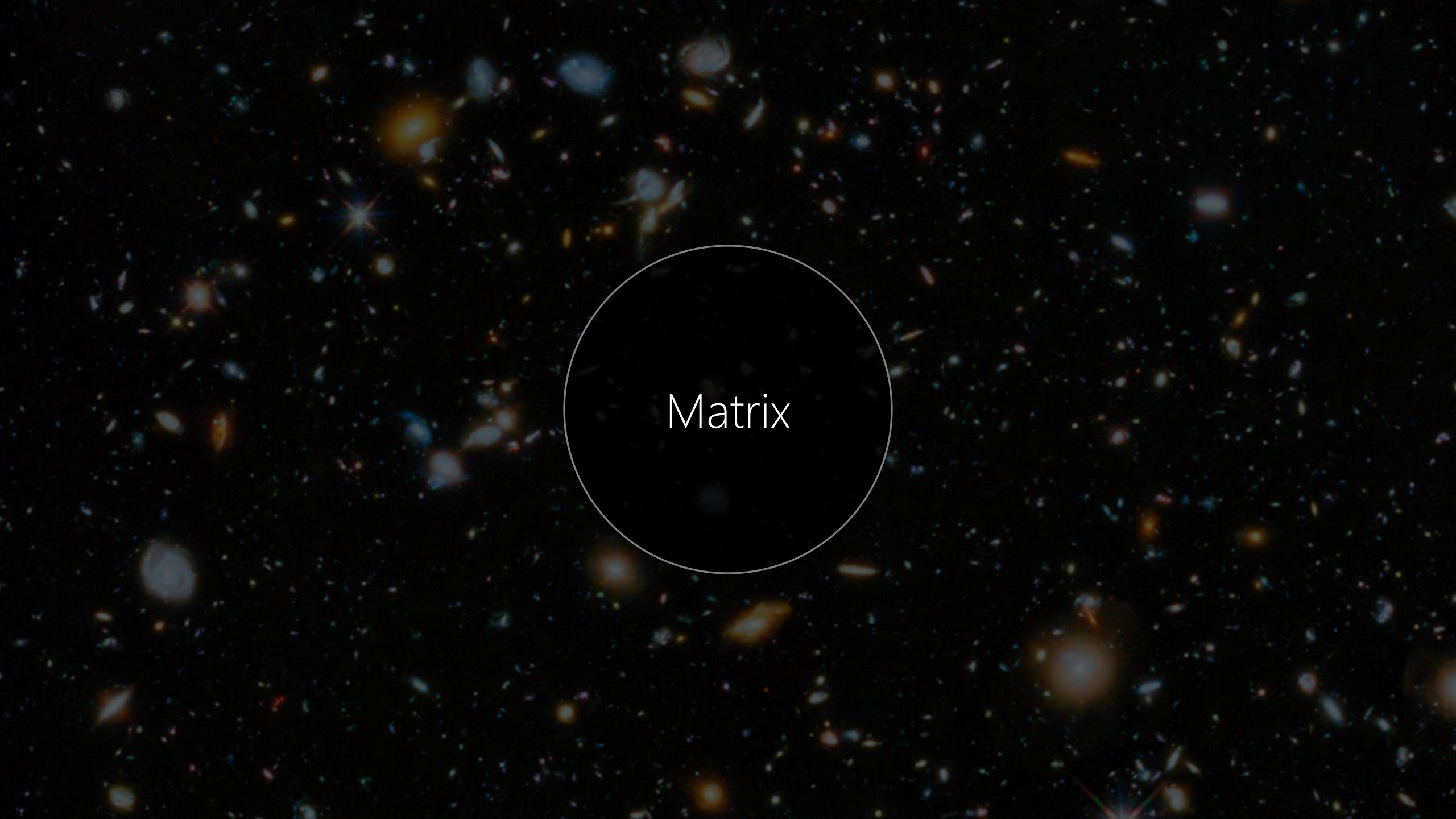


Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
999...99999 + 999...99999 = 1999 ... 9998 => 50,001 digits
Total time = 0.000000



Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
How many iterations:
?
Total time = ?



A deep space image featuring a dense field of galaxies and stars against a black background. The galaxies are mostly blue and yellow, with some showing spiral or elliptical shapes. A large, thin white circle is centered in the image, containing the word "Matrix" in a white, sans-serif font.

Matrix

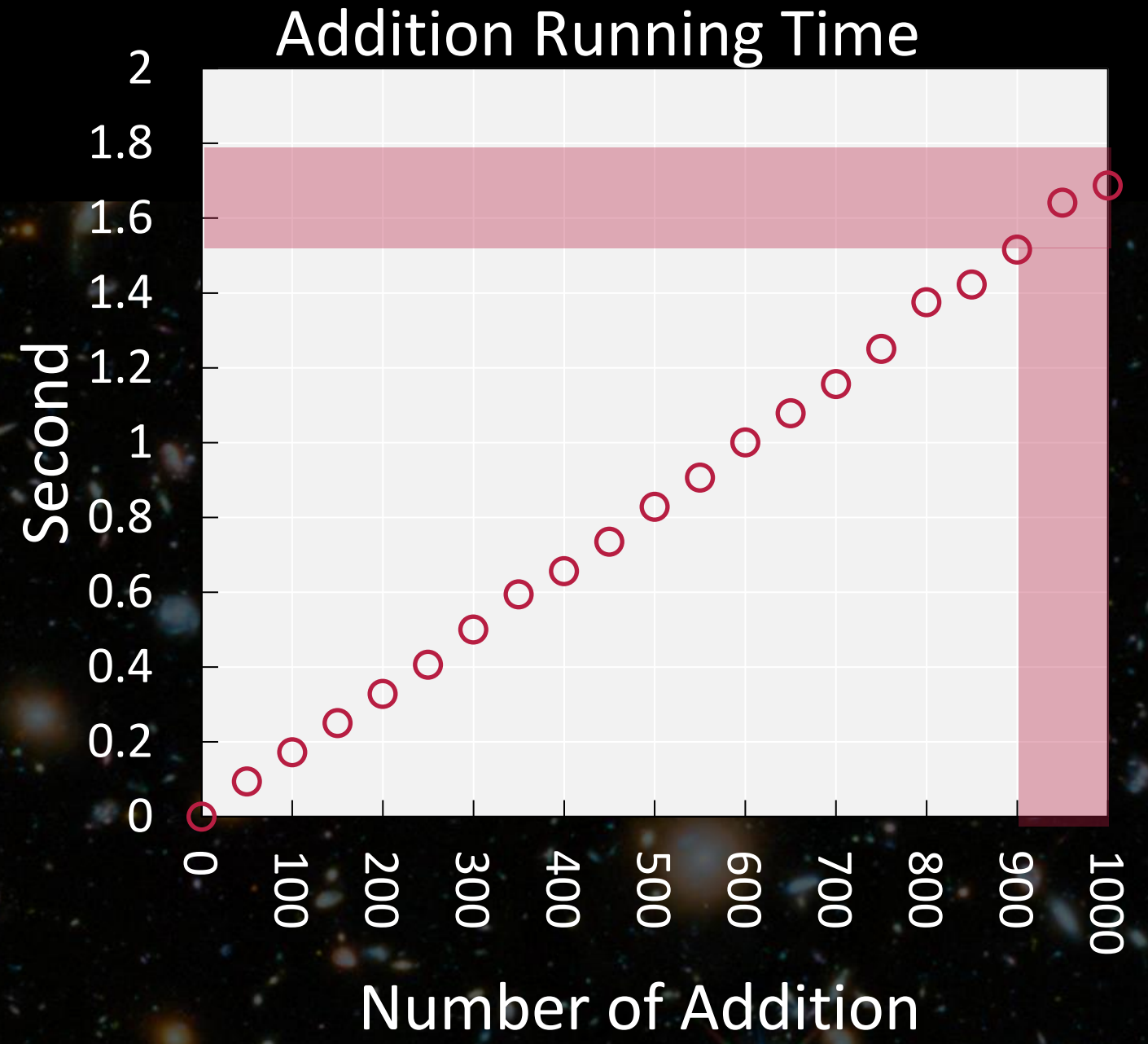
Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
How many iterations:
1
Total time = 0.000000



Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
How many iterations:
100
Total time = ?

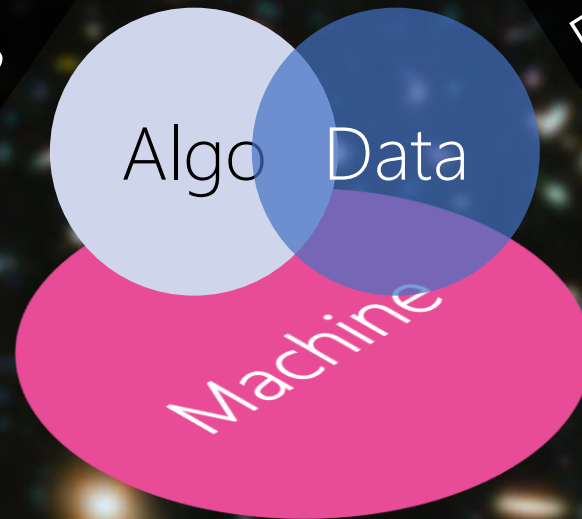


Enter two integers:
999...99999 => 50,000 digits
999...99999 => 50,000 digits
How many iterations:
1,2,3,...,900,..., 1000
Total time =

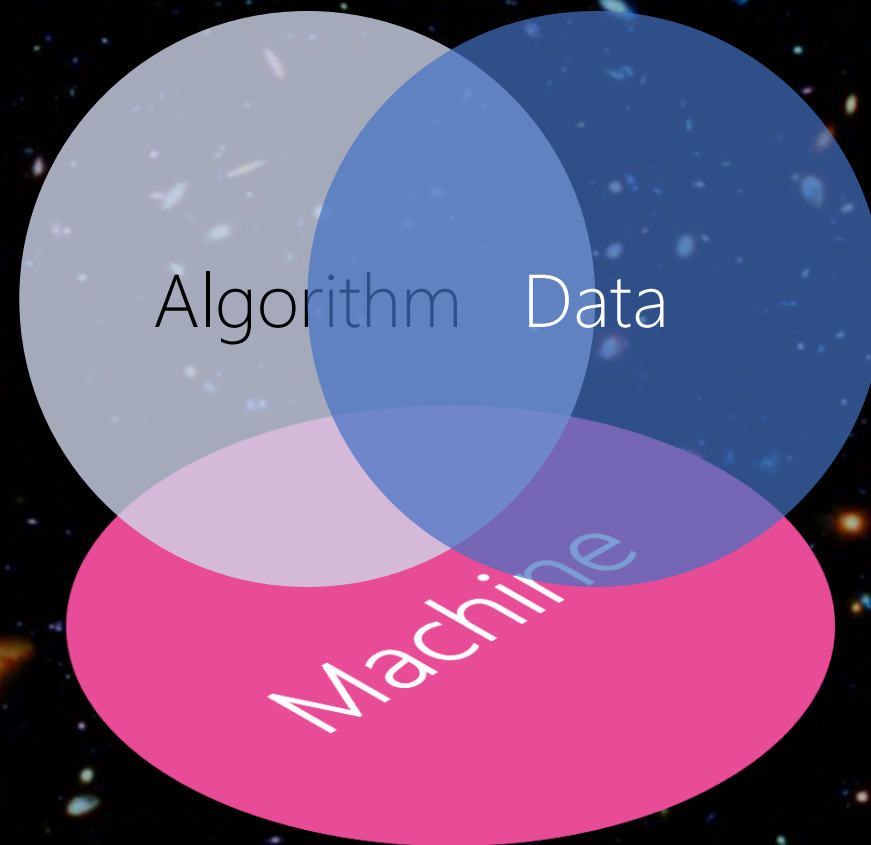
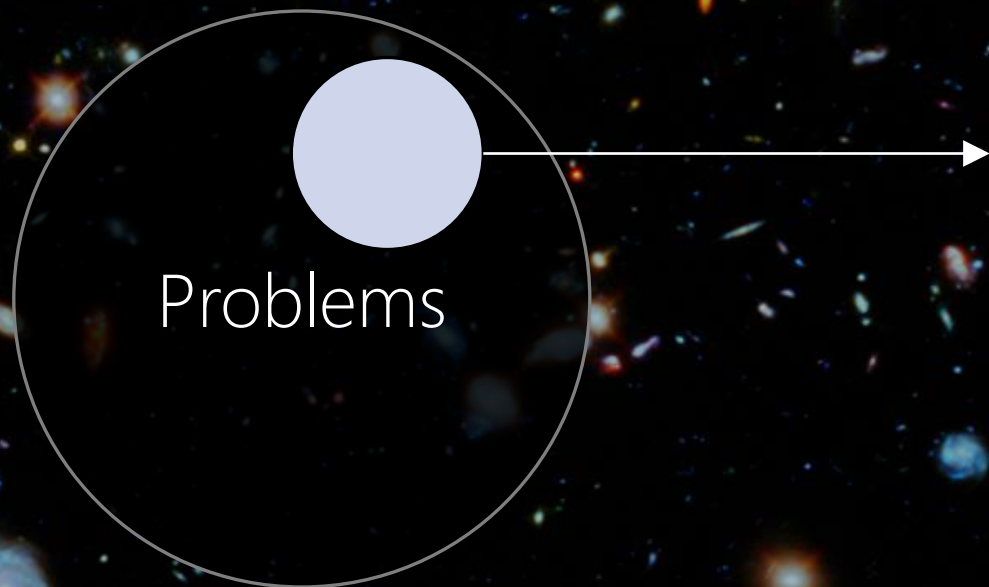


Efficient Algorithm
Less Number of Steps

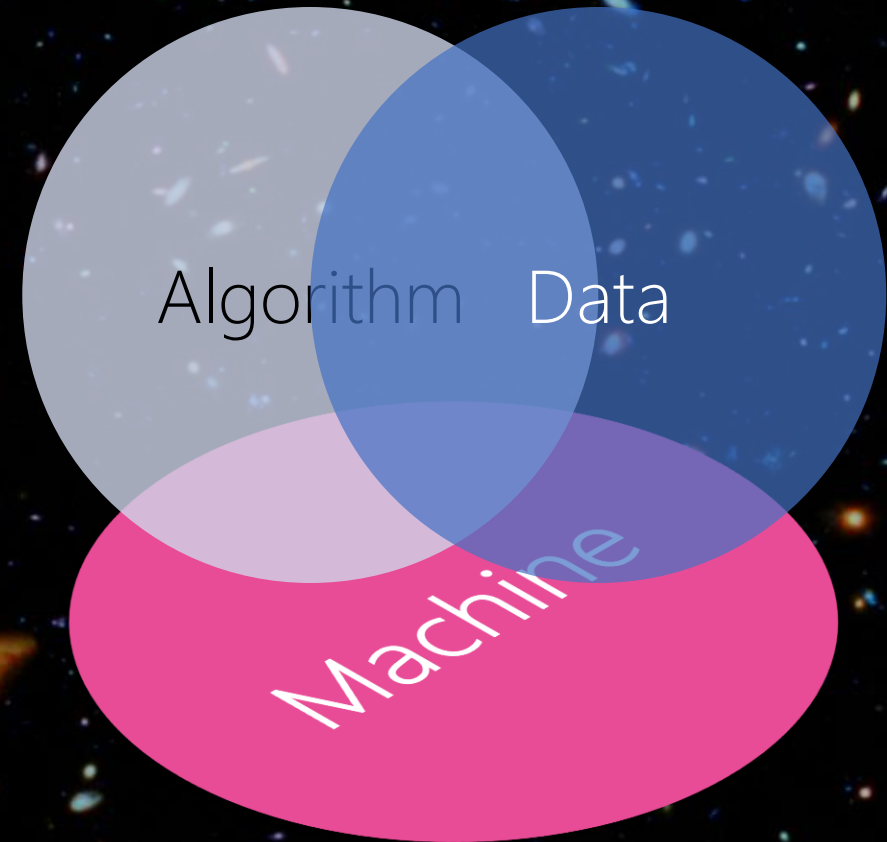
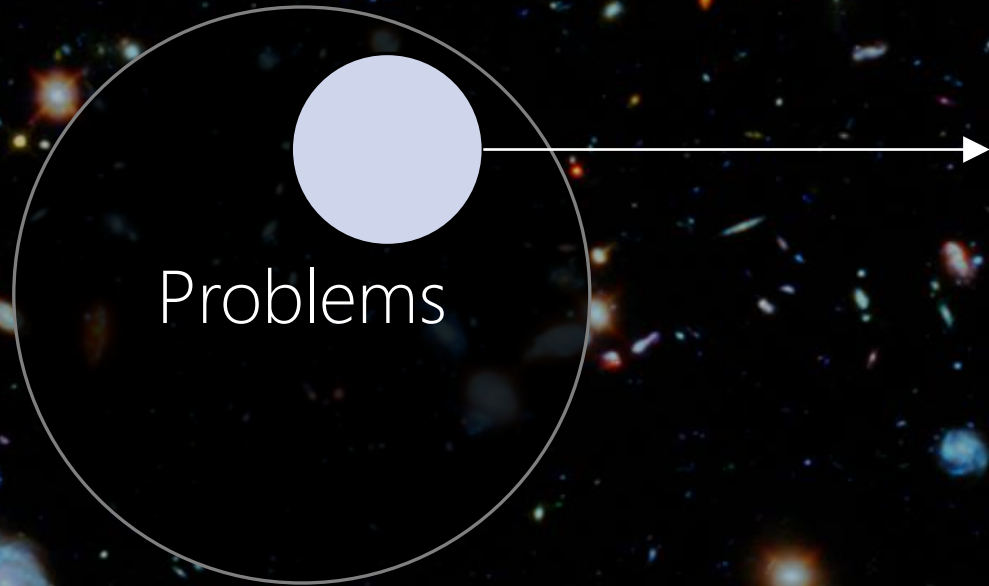
Efficient Data Representation
Smaller Size



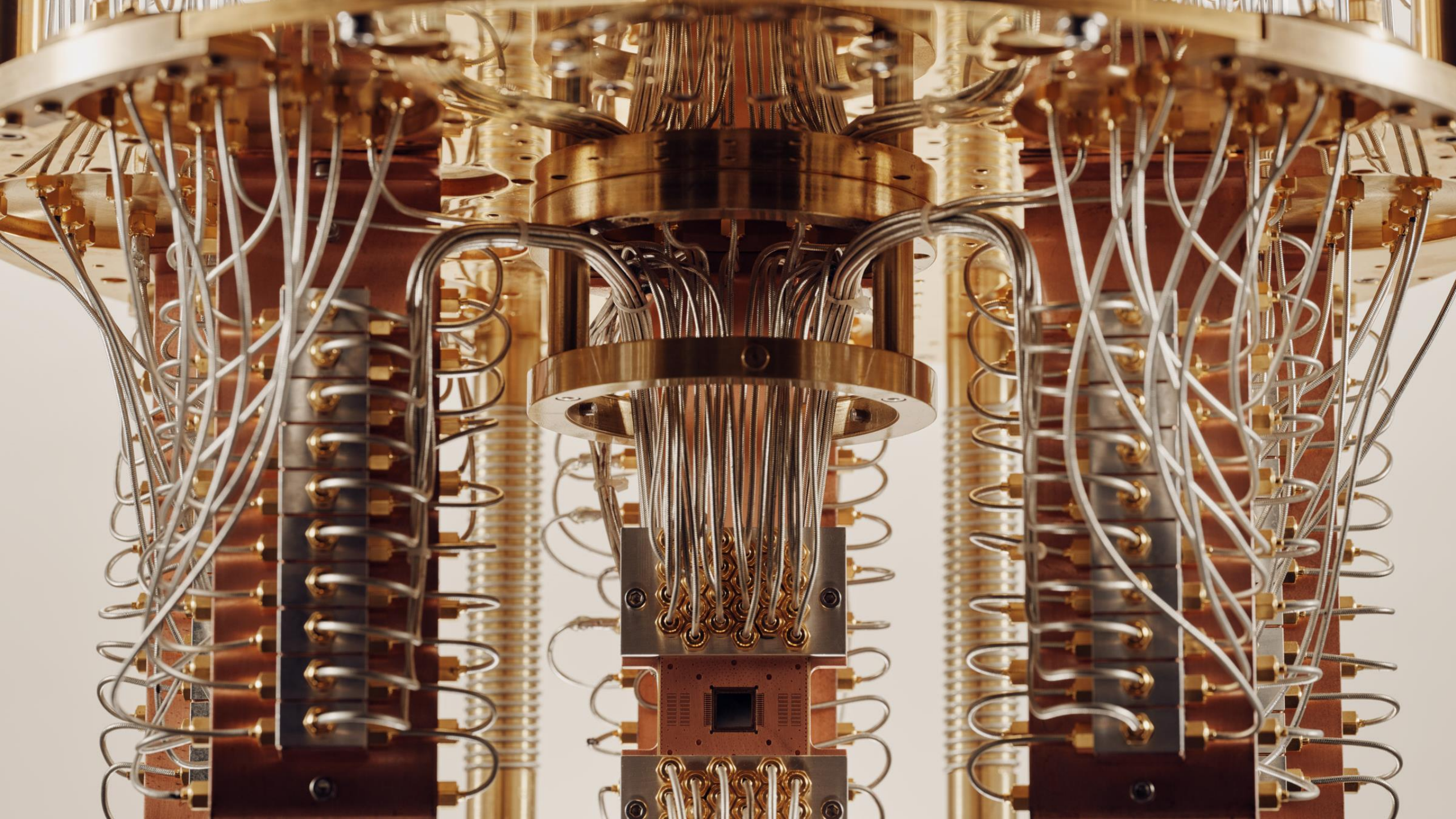
Efficient Machine
Faster Runner

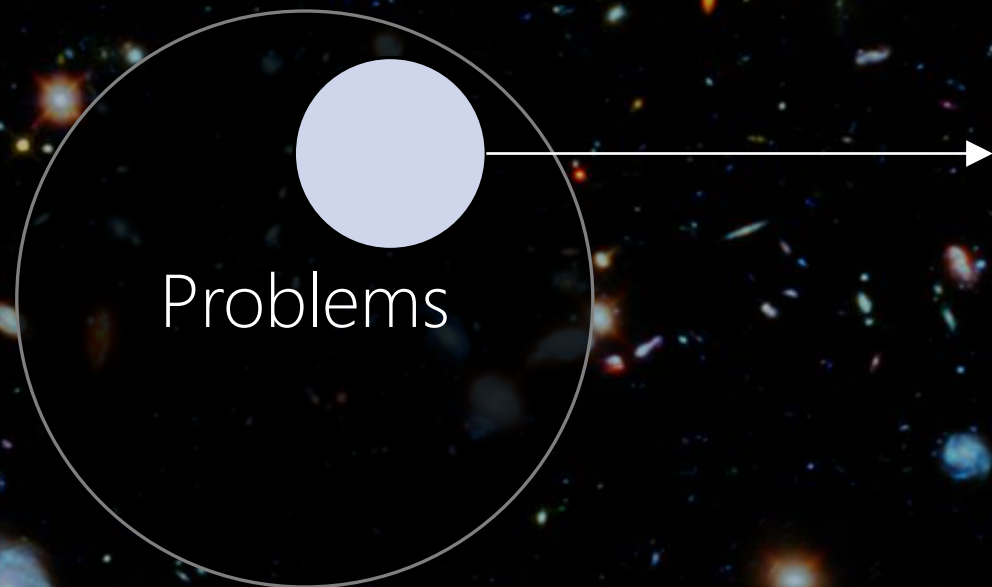


Machine-Friendly Algorithms
Machine-Friendly Data Representation

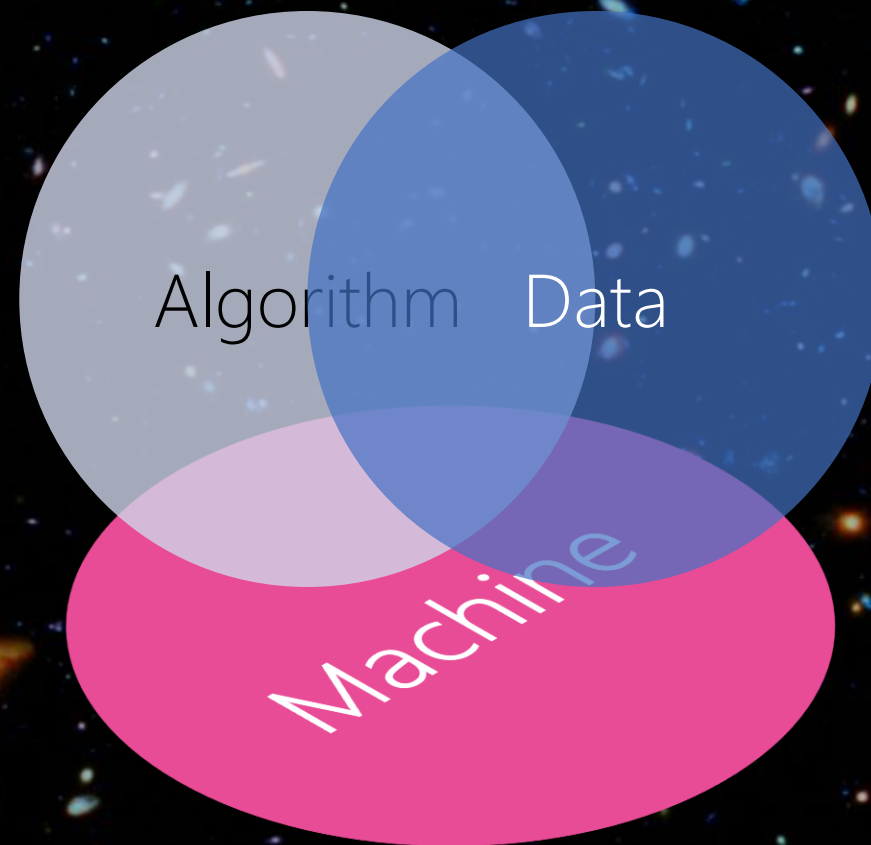


Adding Two Matrix:
Arrays: $O(1) * O(n)$
Linked List: $O(n) * O(n)$





Quantum Computers



Algorithm Design
Algorithm Analysis
Artificial Intelligence (AI)
Machine Learning
Data Mining

Data Structure
File Structure
Database Management Sys.
Data Warehouse
Big Data
Cloud

Algo Data

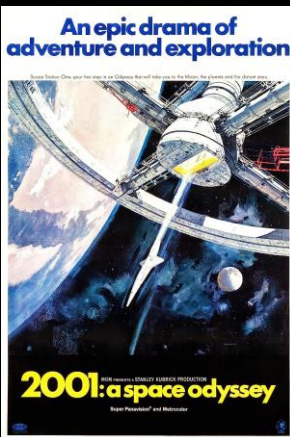
Machine

Digital Design (Logic Circuits)
Computer Architecture
Microprocessor Programming

S2025: A Digital Odyssey

The Ultimate Trip.

"Bone (match) cut"



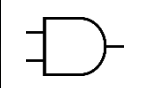
25 Million Years Ago to 2001



Tools: Bone, Computers, Telescopes, Spacecraft, ...

Number Systems $|(12)_{10} \rightarrow (1100)_2$

Number Systems | $(12)_{10} \rightarrow (1100)_2$

Logic Gates | 

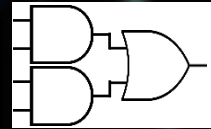
Number Systems $|(12)_{10} \rightarrow (1100)_2$

Logic Gates |



Combinational Logic |

Calculators!

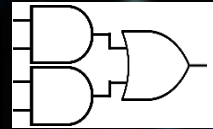


Number Systems $|(12)_{10} \rightarrow (1100)_2$

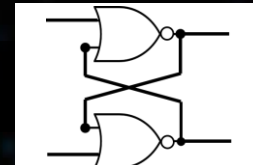
Logic Gates |



Combinational Logic |
Calculators!



Flip-Flop |

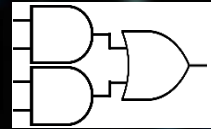


Number Systems $|(12)_{10} \rightarrow (1100)_2$

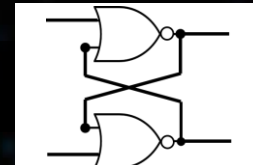
Logic Gates |



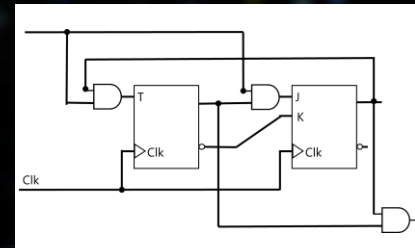
Combinational Logic |

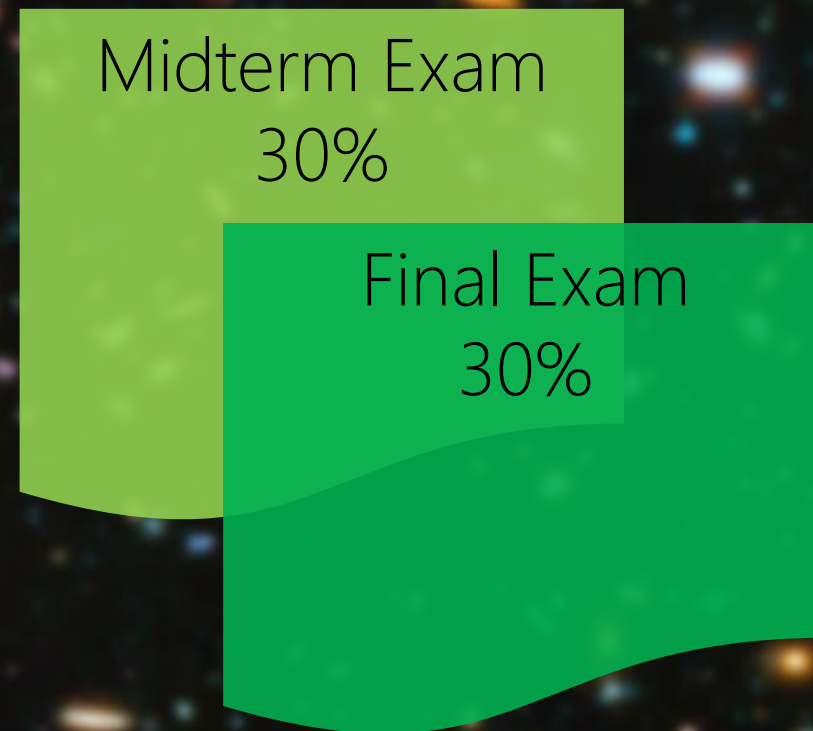
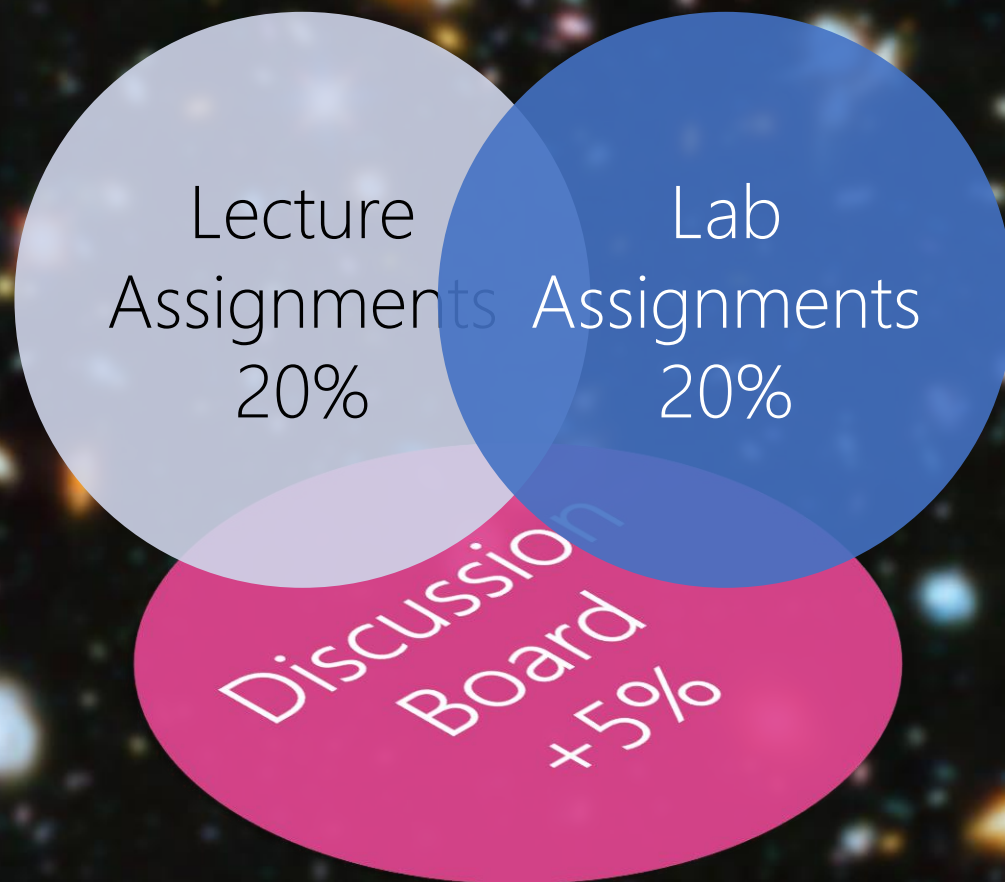


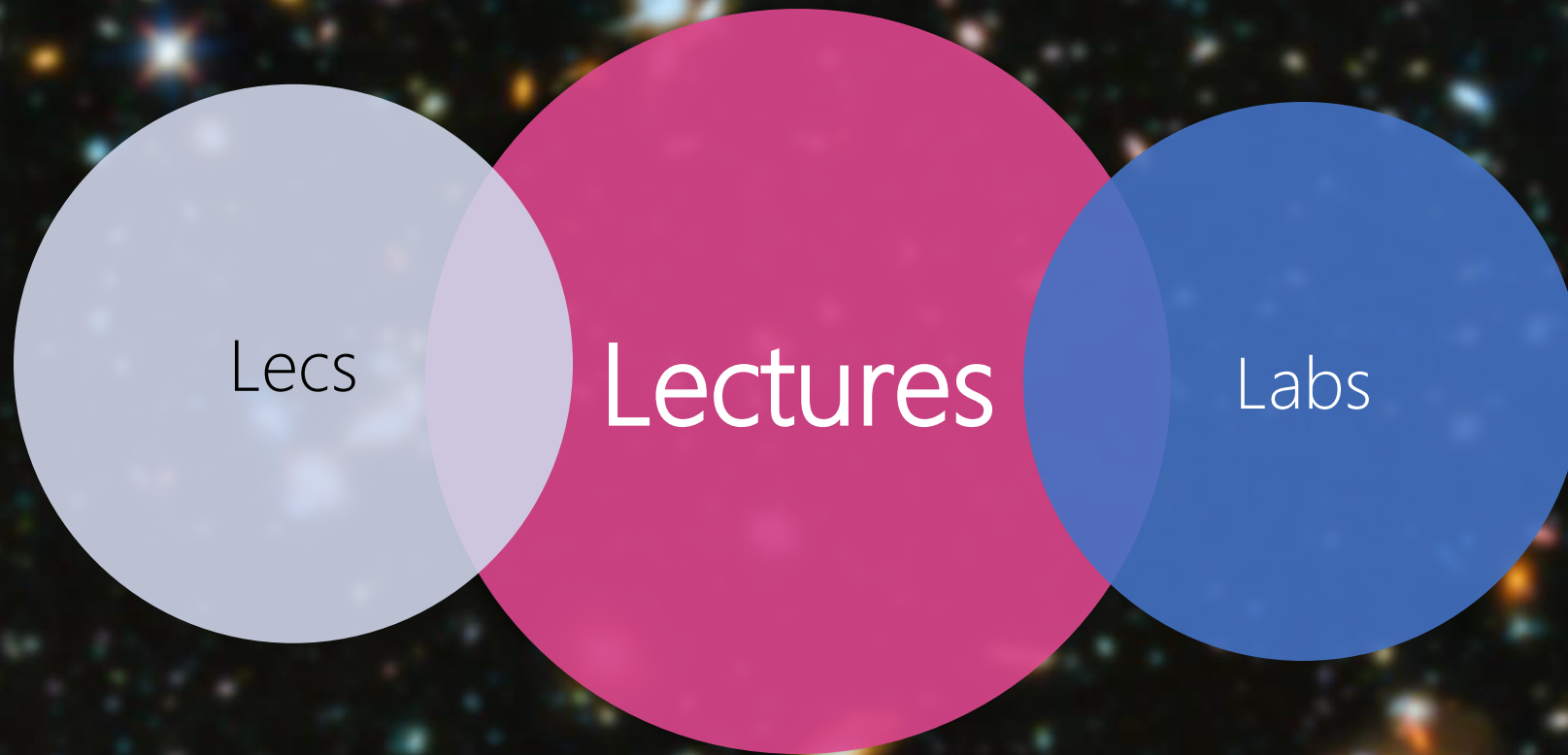
Flip-Flop |



Sequential Logic |
Computers







Some topics are not covered in the lecture, but either in Labs or Lecs

- Change of classroom and labrooms
- Assignments
 - For marking and any related issue, foremost reach out to your lab instructor, not the marker. Then me!
 - For opening another submissions
 - Pay attention to naming of files!
 - Some topics are not covered in the lecture, but in labs or lecture assignments
 - Lab may have dependencies.
 - No key to the labs!
 - Key to the assignments based on highest selected assignments
 - Questions about the lab or lec in discussion board.
 - Subscribe to the forums to receive emails when somebody put sth
 - Academic Integrity (plagiarism)
- Keys to the exams (midterm and final)

- Blackboard is unavailable every weekdays (Monday to Friday) from 5 AM to 6 AM for regular maintenance!
- Change of classroom and labrooms
- Assignments
 - For marking and any related issue, foremost reach out to your lab instructor, not the marker. Then me!
 - For opening another submissions
 - Pay attention to version of files

SLIDES ARE NOT LIKE THIS

- No key to the labs.
 - Key to the assignments based on highest selected assignments
 - Questions about the lab or lec in discussion board.
 - Subscribe to the forums to receive emails when somebody put sth
- Keys to the exams (midterm and final)

QUALITY

4.0

COMP2650

Dec 11th, 2024

For Credit: **Yes** Attendance: **Not Mandatory** Would Take Again: **Yes** Grade: **B**

Textbook: **Yes**

DIFFICULTY

4.0

definitely a great prof who was extremely understanding and accommodating to all scenarios his lectures can be hard to follow and his but tests are difficult however he gives you a lot of chances to do well and get good marks. Genuinely an amazing person who went above and beyond to be accessible and help students succeed

EXTRA CREDIT

CLEAR GRADING CRITERIA

CARING

Helpful  0  0



QUALITY

2.0

COMP2560

Dec 14

For Credit: **Yes** Attendance: **Not Mandatory** Would Take Again: **Yes** Grade: **B+**

Textbook: **Yes**

DIFFICULTY

2.0

you need to read to much from outside, lecture slide are not useful especially in exam

GET READY TO READ

CLEAR GRADING CRITERIA

RESPECTED

Helpful  0  0

Are robots becoming more human or humans becoming more robotic?
<https://www.youtube.com/watch?v=YR5ApYxkU-U>

QUALITY

2.0

COMP2650

Mar 20th, 2025

For Credit: **Yes** Attendance: **Not Mandatory** Grade: **Incomplete** Textbook: **Yes**

Fani inspired me to not be another brick in the **wall**. Besides that, he cannot teach.

DIFFICULTY

4.0

INSPIRATIONAL

Helpful  0  0



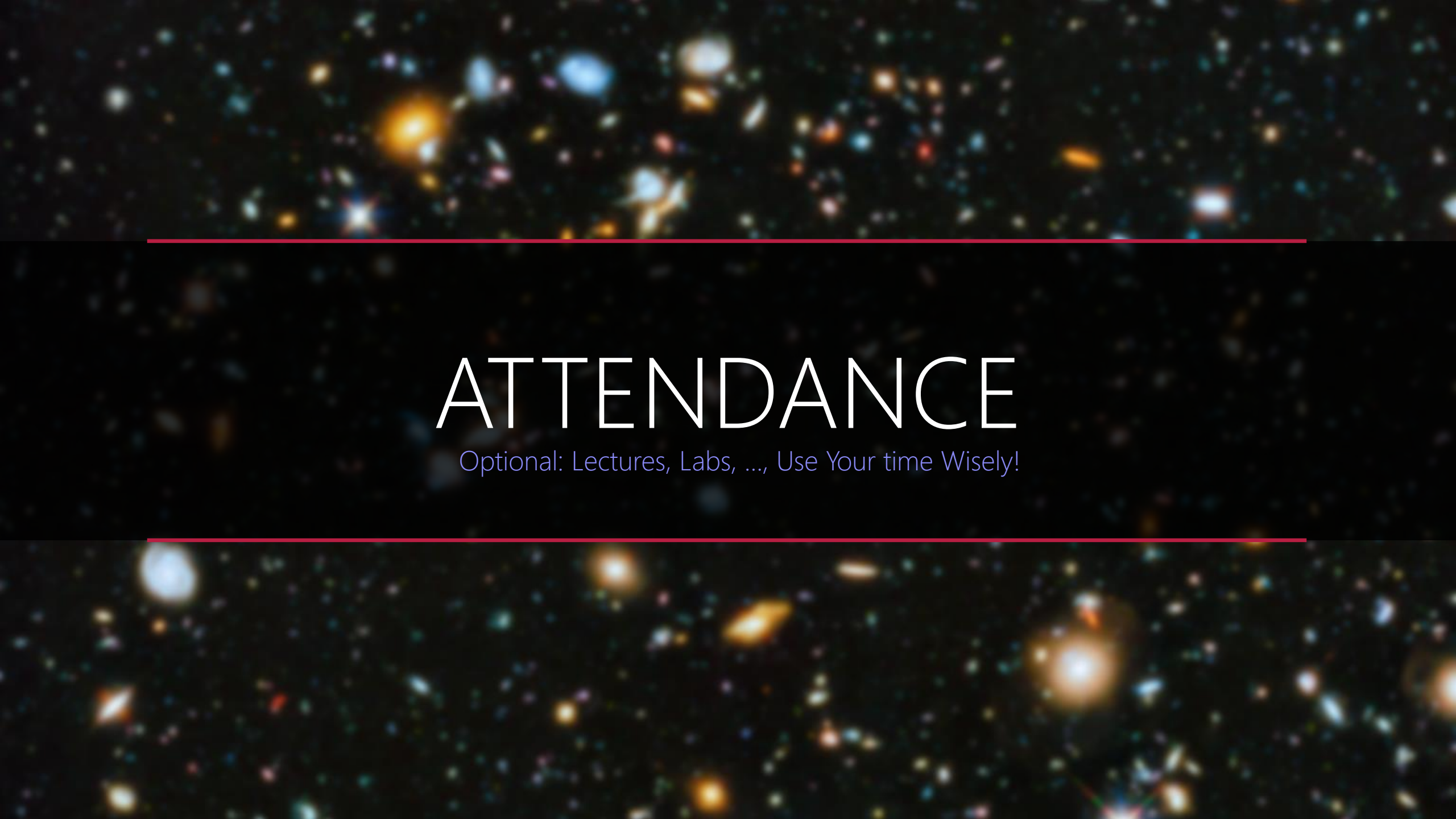
Slides are not books, summaries, ...
Slides are not for last night before exam
Slides are to be presented! Lecture = Slide + Talk

NOTETAKERS NEEDED

**VOLUNTEER
TO BE A
NOTETAKER**

We are looking
for a volunteer
notetaker for this
class. If you would
like to volunteer
please email
notetaker@uwindSOR.ca



A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, diffuse clouds and others as more compact, elongated structures. The colors range from deep blues and purples to bright yellows and oranges, set against a dark, star-filled sky.

ATTENDANCE

Optional: Lectures, Labs, ..., Use Your time Wisely!

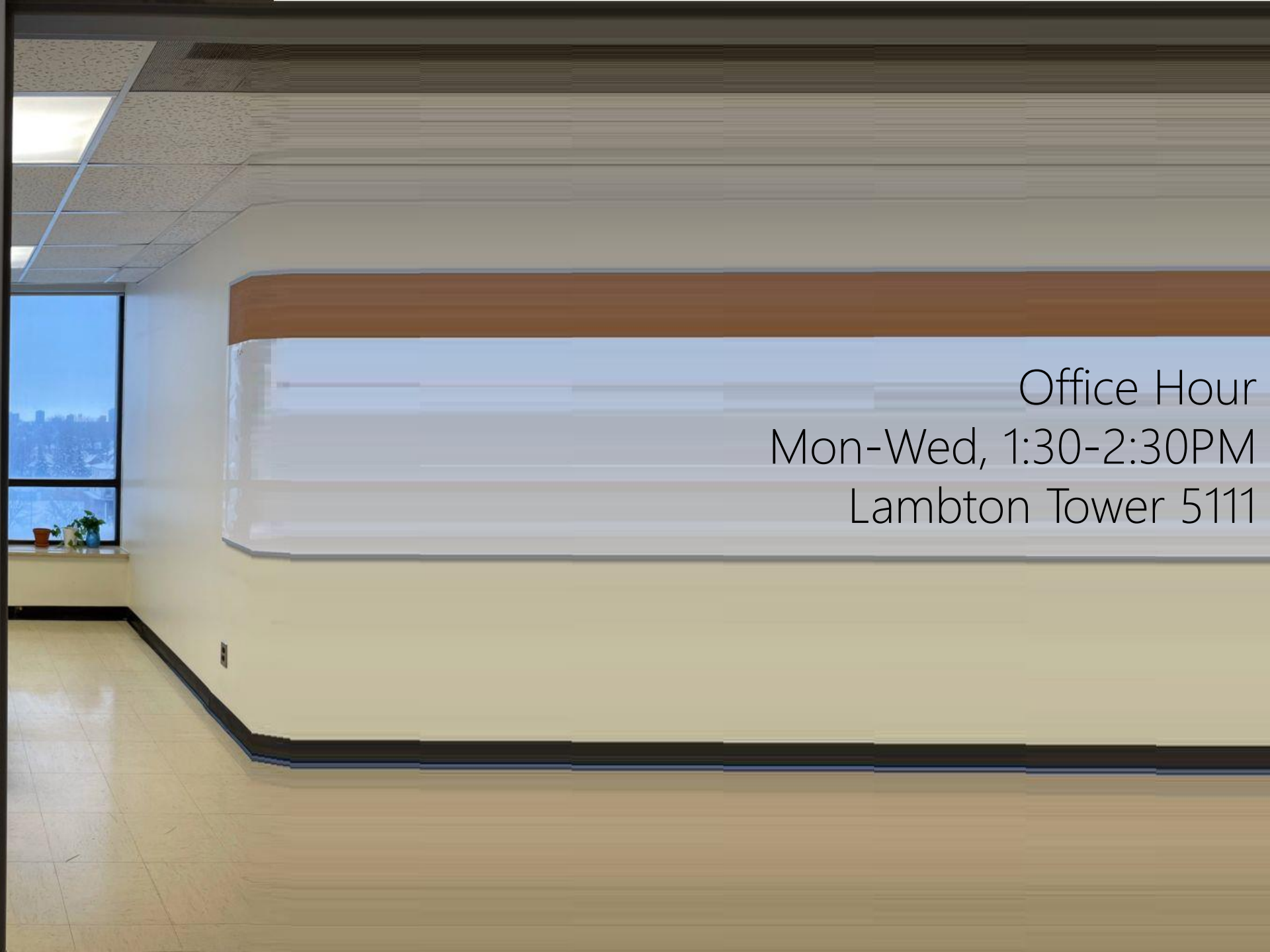


Hossein Fani, PhD

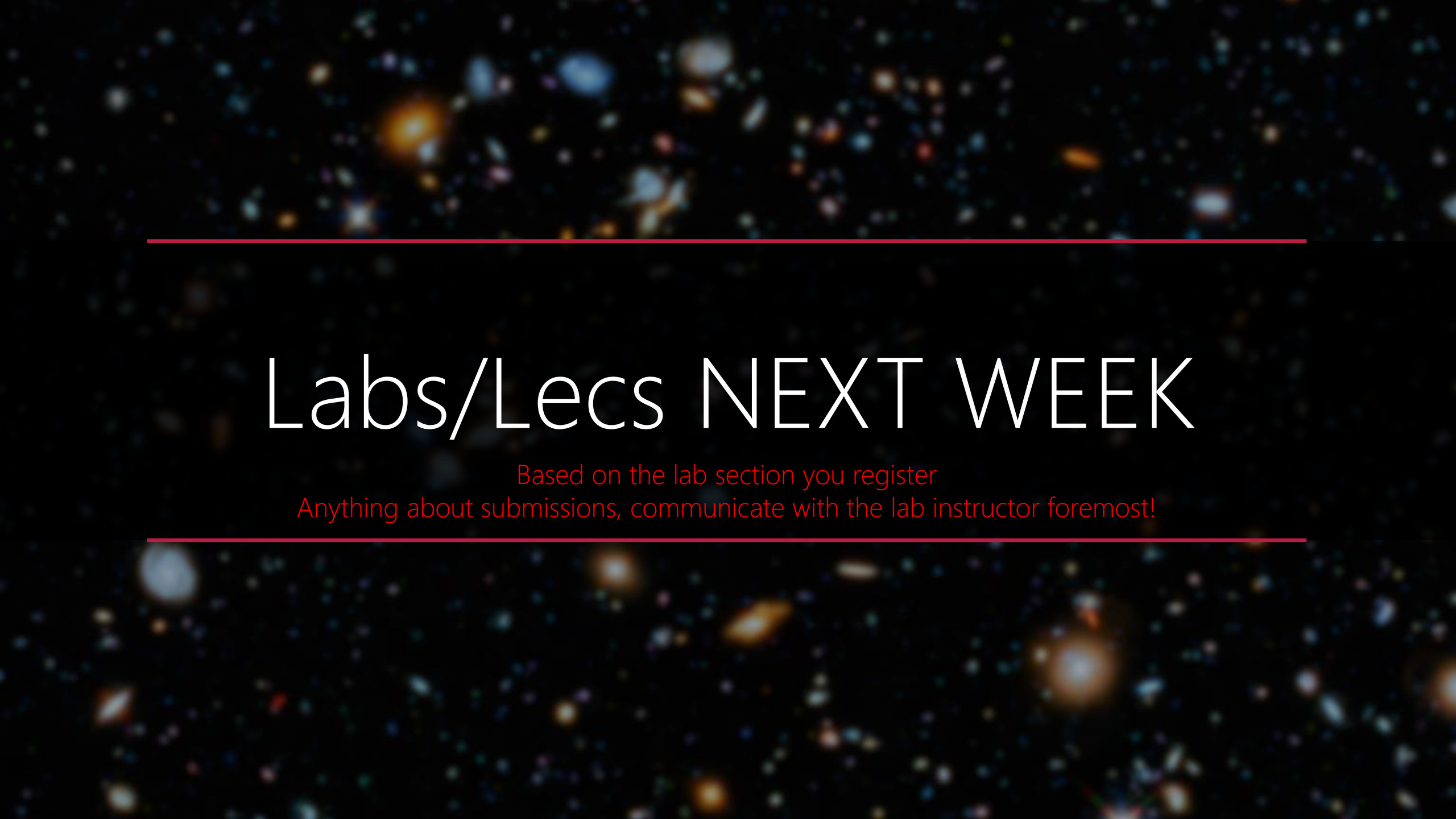
Assistant Professor, School of Computer Science, Faculty of Science, University of Windsor
Room 5111, Lambton Tower

hfani@uwindsor.ca

cs.uwindsor.ca/~hfani



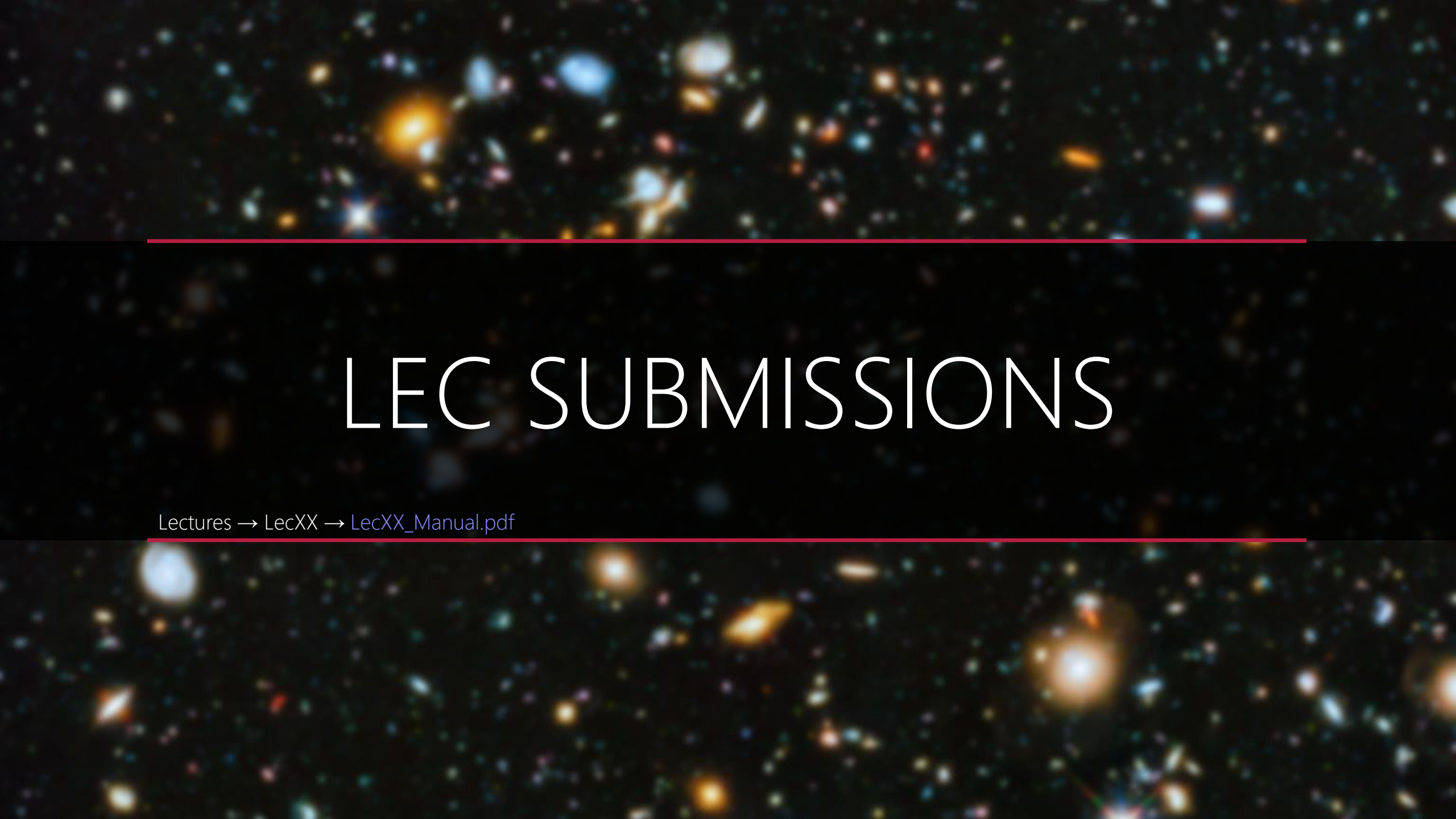
Office Hour
Mon-Wed, 1:30-2:30PM
Lambton Tower 5111

The background of the slide is a deep space image showing numerous galaxies in various colors (blue, orange, white) against a black sky. A solid red horizontal line spans the width of the slide, positioned above the main title.

Labs/Lecs NEXT WEEK

Based on the lab section you register
Anything about submissions, communicate with the lab instructor foremost!

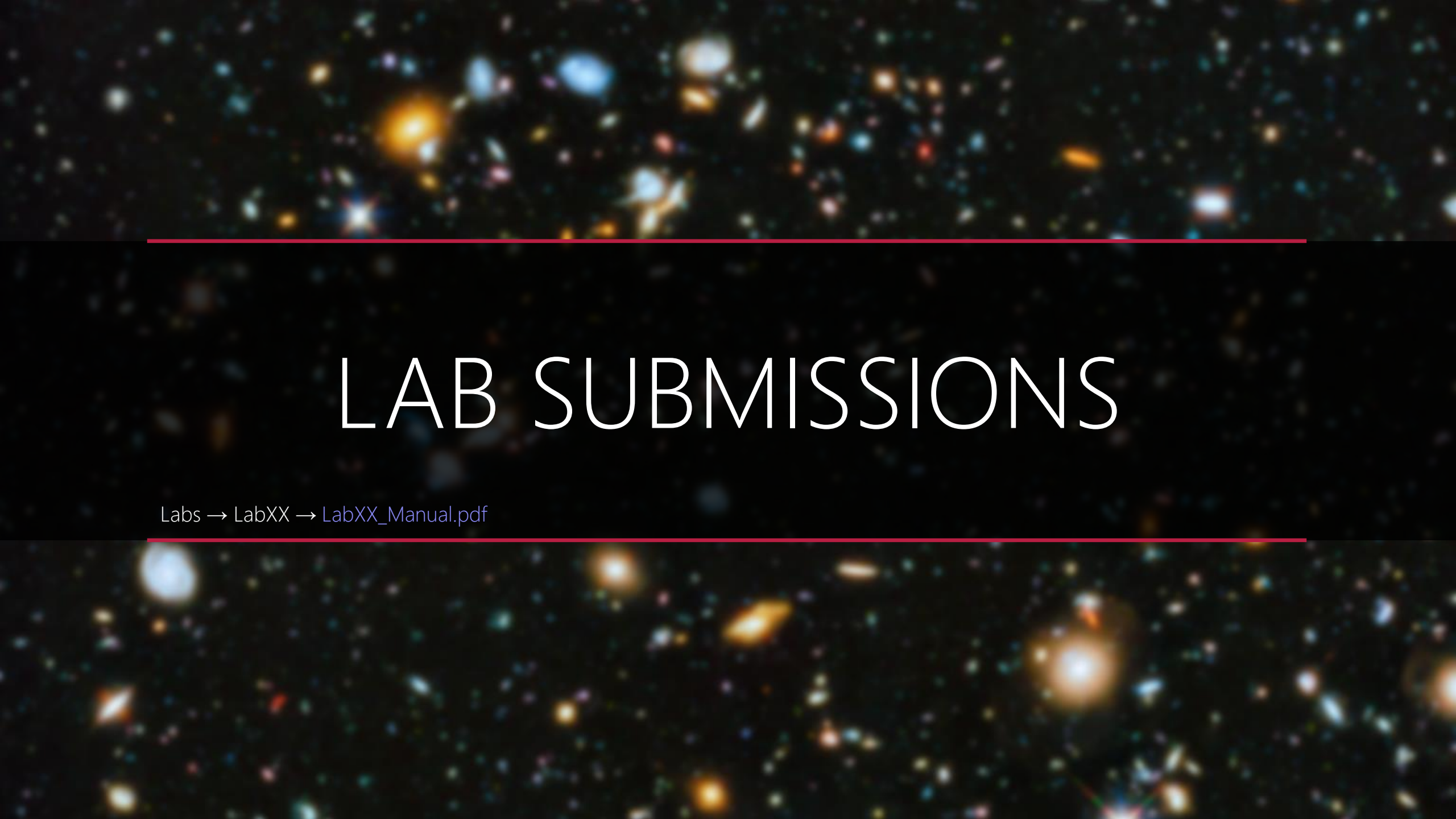
A solid red horizontal line spans the width of the slide, positioned below the text.

A deep-field astronomical image showing a vast number of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, fuzzy blobs and others as more distinct, elongated structures. A thin red horizontal line is positioned above the title text.

LEC SUBMISSIONS

Lectures → LecXX → [LecXX_Manual.pdf](#)

A continuation of the cosmic background image from the top, showing a dense field of galaxies. A thin red horizontal line is positioned below the navigation text.



LAB SUBMISSIONS

Labs → LabXX → [LabXX_Manual.pdf](#)

Lab instructors' photos will be updated



Jaber Al Siam (siamj@uwindsor.ca)

#51: Monday 5:30PM-6:50PM, Odette Building B06

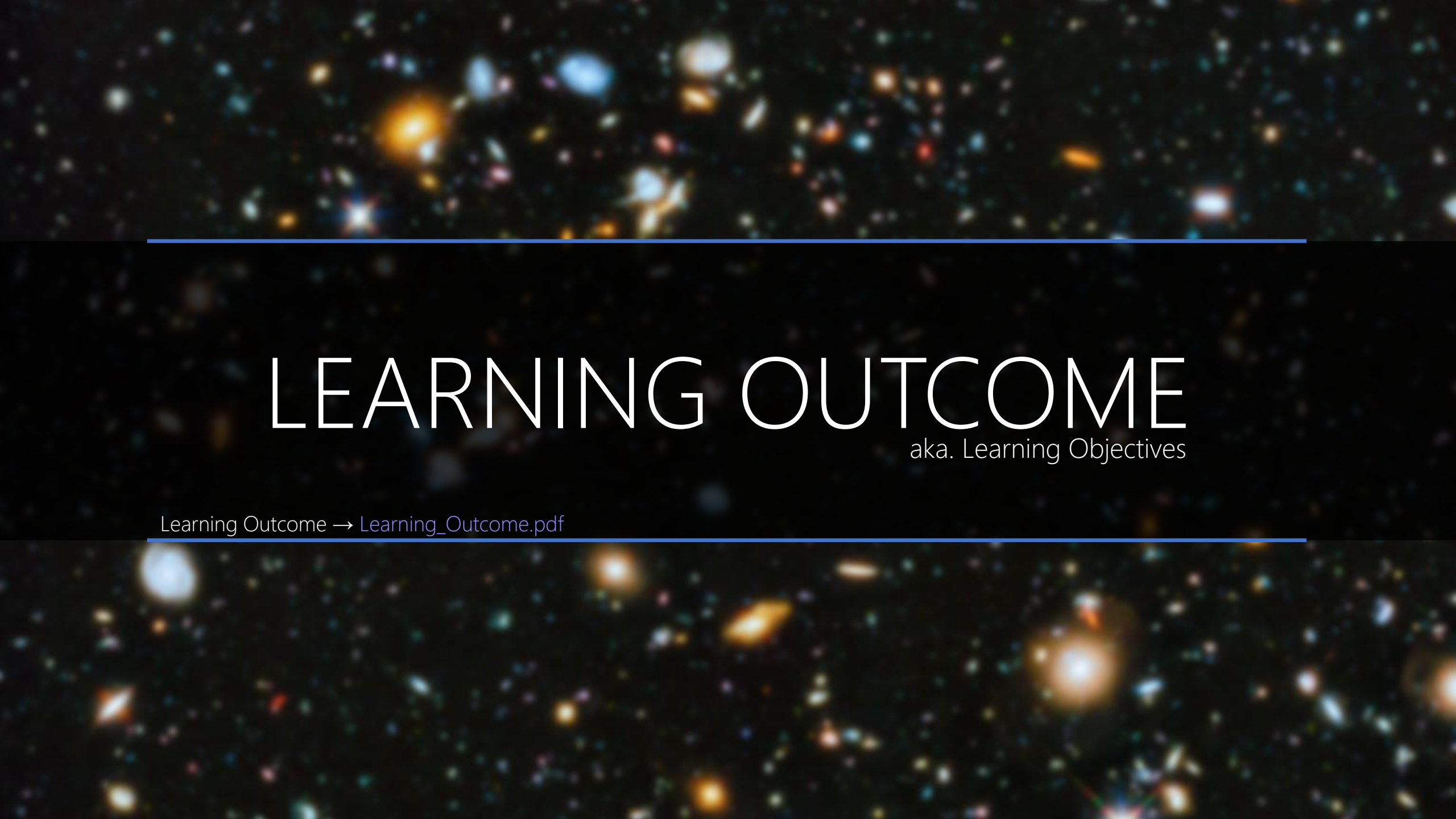


Rustem Izmailov (izmailor@uwindsor.ca)

#52: Wednesday 5:30PM-6:50PM, Odette Building B06

The background of the entire image is a deep space photograph showing a vast field of galaxies and stars. The galaxies are in various stages of evolution, some appearing as bright, smooth ellipsoids and others as more complex, irregular shapes. The stars are small, distant points of light, many of which are multi-colored, suggesting they have been filtered to show different spectral components. The overall scene is a dense, colorful mosaic of celestial objects against the blackness of space.

BRIGHTSPACE TOUR

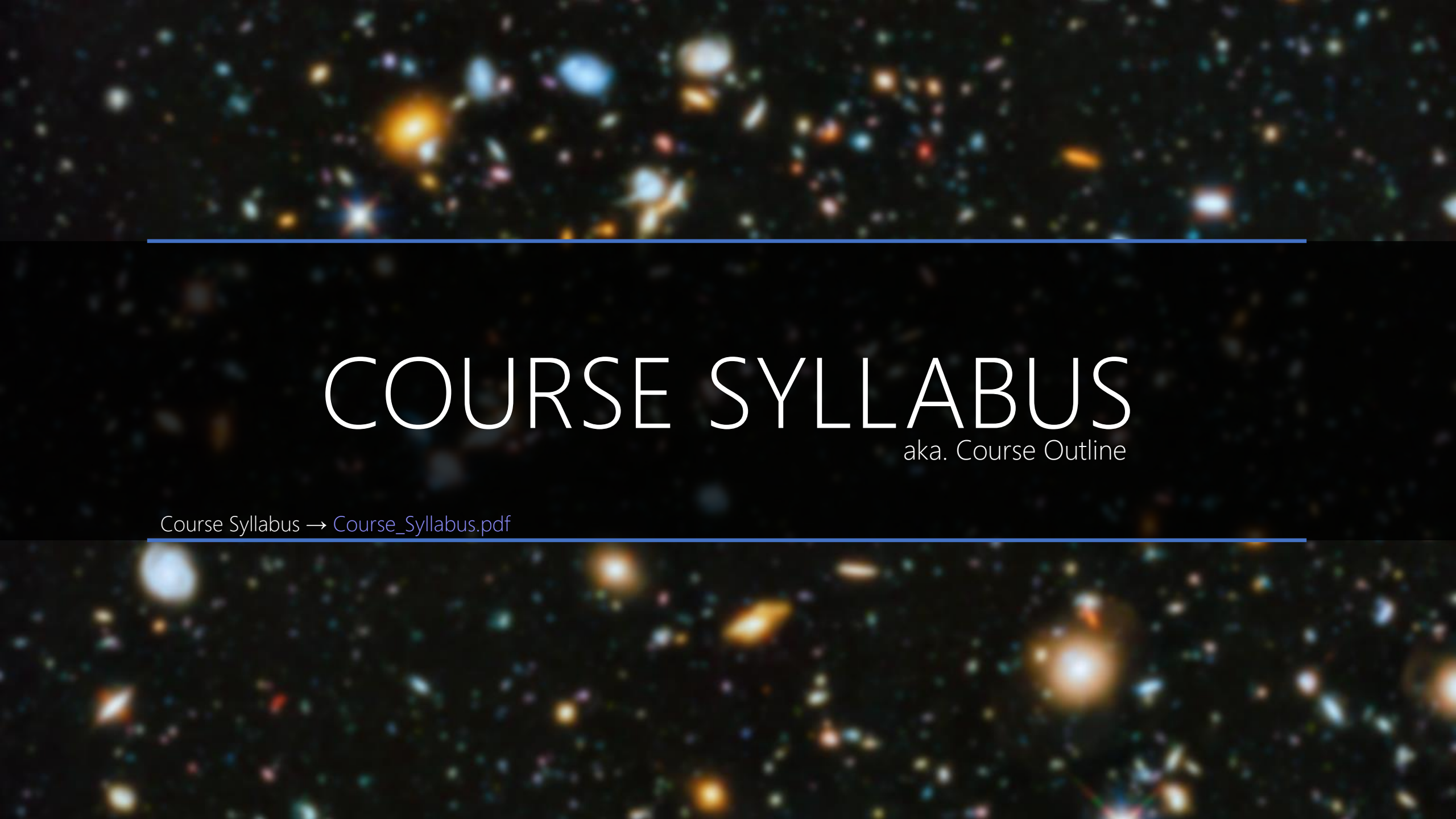
A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a dark space. A horizontal blue line is positioned above the main title.

LEARNING OUTCOME

aka. Learning Objectives

Learning Outcome → [Learning_Outcome.pdf](#)

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a dark space. A horizontal blue line is positioned below the text.

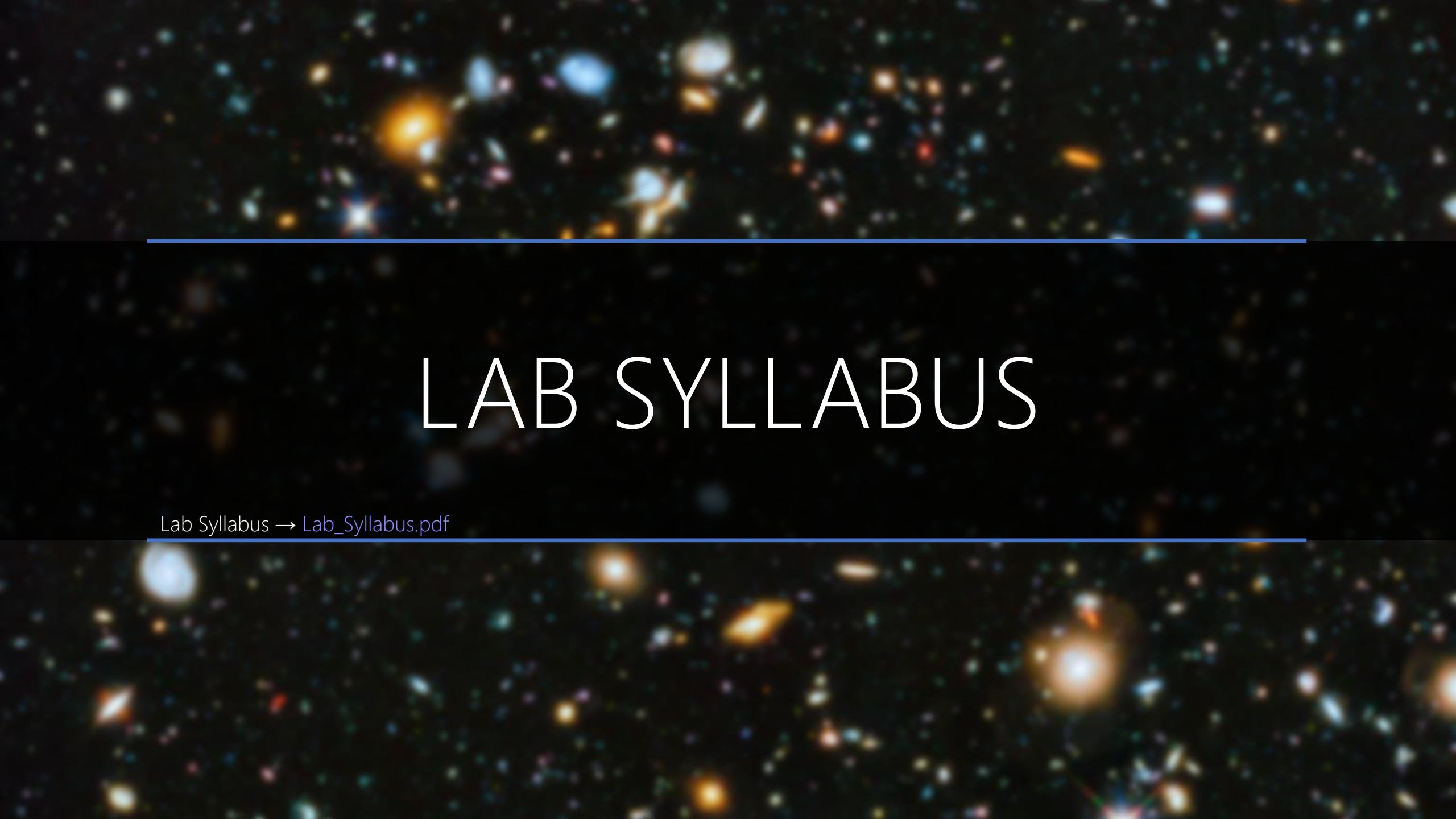
A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, red) against a black space. A horizontal blue line is positioned above the main title.

COURSE SYLLABUS

aka. Course Outline

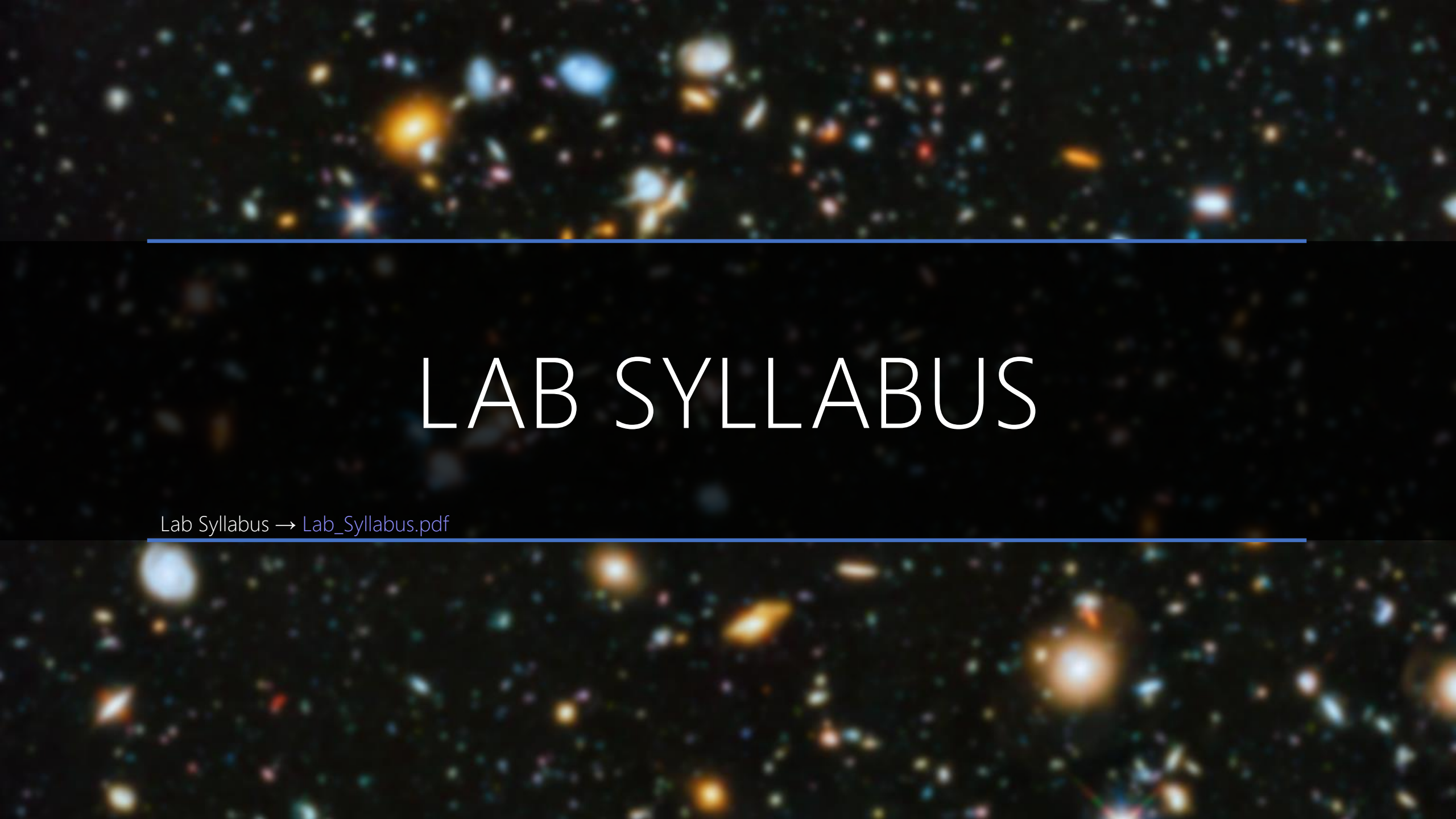
Course Syllabus → [Course_Syllabus.pdf](#)

A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, red) against a black space. A horizontal blue line is positioned below the text.

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, diffuse clouds and others as more compact, point-like sources. A thin blue horizontal line is positioned above the title.

LAB SYLLABUS

Lab Syllabus → [Lab_Syllabus.pdf](#)

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, diffuse clouds and others as more compact, point-like sources. A thin blue horizontal line is positioned below the text.

master 3 Branches 2 Tags

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Labs	Winter 2022, up to W12	3 years ago	
Lectures	w12-w13. end of w2022	3 years ago	
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.gitattributes	Initial commit	5 years ago	
.gitignore	no message	4 years ago	
Book.jpg	Initial Commit	5 years ago	
Course_Syllabus.docx	Winter 2022, up to W12	3 years ago	
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Schedule_Timeline.xlsx	Winter 2022, up to W12	3 years ago	

About

Materials for the Computer Science course, Digital Design (Logic Circuits)

- counter
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- decoding
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- logic-gates
- boolean-algebra
- multiplexer
- adder
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- registers
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- number-systems
- subtractor
- logic-circuits
- programmable-logic

- Readme
- View license
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- 7 stars
- 1 watching
- 4 forks
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Releases 2

Winter2021 at UWindsor Latest on Jan 10, 2022

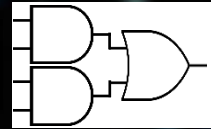
+ 1 release

Number Systems $|(12)_{10} \rightarrow (1100)_2$

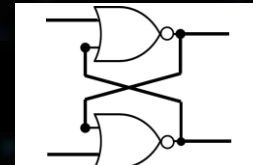
Logic Gates |



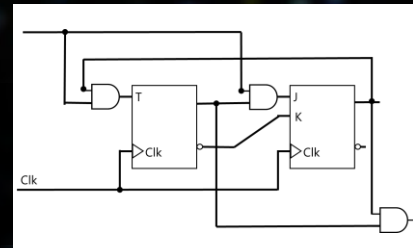
Combinational Logic |



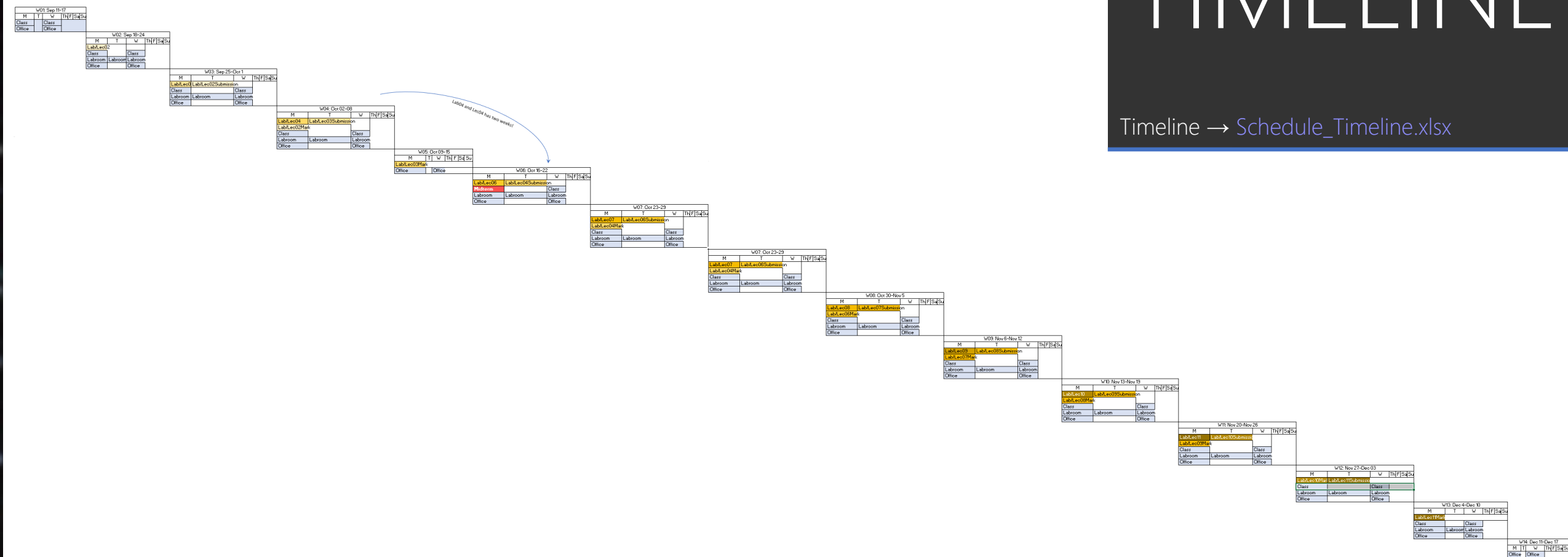
Flip-Flop |



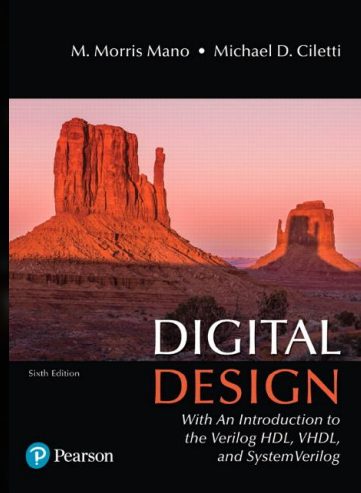
Sequential Logic |
Computers



Eastern Time	SUN	MON	TUE	WED	THU	FRI	SAT
0:00		Marks for Submissions Manuals for Submissions	Deadline for Submissions				
8:30		Lecture A Instructor: Hossein Human Kinetics Building 140		Lecture B Instructor: Hossein Human Kinetics Building 140			
10:00							
11:30		Lab Section 51 Lab Instructor: Zahra Dillon Hall 365	Lab Section 55 Lab Instructor: Zahra-Roonak Odette Building 110	Lab Section 52 Lab Instructor: Roonak Dillon Hall 354			
13:00		Lab Section 53 Lab Instructor: Zahra Chrysler Hall North G100		Lab Section 54 Lab Instructor: Roonak Chrysler Hall North G100			
14:20							
15:00		Office A Instructor: Hossein Lambton Tower S111		Office B Instructor: Hossein Lambton Tower S111			
16:00							



Number Systems $|(12)_{10} \rightarrow (1100)_2$



Chapter 1

Digital Systems and Binary Numbers



The Bonnie Situation, Pulp Fiction (1994), Quentin Tarantino